

Research article

E-SF2I: Dynamics modeling and multi-level verification of Online Social Network rumor propagation driven by emotional differentiation and dual-immunity-state coupling

Chenquan Gan^{a,b}, Hongming Chen^a, Qingyi Zhu^b, Rui Tang^c, Deepak Kumar Jain^{d,e,*}, Liping Feng^f, Salvador García^g

^a School of Communications and Information Engineering, Chongqing University of Posts and Telecommunications, Chongqing 400065, China

^b School of Cyber Security and Information Law, Chongqing University of Posts and Telecommunications, Chongqing 400065, China

^c School of Cyber Science and Engineering, Sichuan University, Chengdu 610065, China

^d Key Laboratory of Intelligent Control and Optimization for Industrial Equipment of Ministry of Education, Dalian University of Technology, Dalian 116024, China

^e Symbiosis Institute of Technology, Symbiosis International University, Pune 412115, India

^f Computer science department, Xinzhou Normal University, Xinzhou 034000, China

^g Department of Computer Science and Artificial Intelligence, Andalusian Research Institute in Data Science and Computational Intelligence, University of Granada, Granada 18071, Spain

ARTICLE INFO

Keywords:

Online Social Networks

Emotional differentiation

Dual immunity state

Propagation dynamics model

Multi-level verification

ABSTRACT

The open architecture of Online Social Networks (OSNs) facilitates information sharing but also exacerbates the risk of rumor propagation. In response to the challenge that existing propagation models struggle to quantify the synergistic effects of user emotional polarization and temporary neutral behavior, this paper proposes a novel E-SF2I (Emotion-Susceptible-Forwarding-Temporary Immunity-Permanent Immunity) rumor propagation dynamics model. In terms of theoretical modeling, a differential dynamic system with time-varying characteristics is constructed by coupling the emotion-driven forwarding mechanism with dual immunity state transitions. Based on the Lyapunov stability theory, the system equilibrium and its global asymptotic stability are rigorously demonstrated. A rumor propagation simulation algorithm based on the E-SF2I framework and a multi-level verification are further developed to verify the practical application value of the proposed model. Initially, the theoretical results derived from the model are validated using real network structure datasets (Facebook, Twitter, P2P). Subsequently, the validity of the proposed model is tested using 13 real rumor events on the Twitter dataset. Finally, the proposed model is compared with traditional and new models for three representative real rumor events. The results show that the proposed model achieves higher accuracy in predicting the propagation trends of rumor events.

1. Introduction

Online Social Networks (OSNs), a classic manifestation of the digital social framework, are co-formed by nodes of social entities and their interactive connections (Liu, He, Yang, & Liu, 2019). With the booming popularity of social media platforms, OSNs have transformed into a powerful platform for immediate information exchange (Hu et al., 2023). Their open and decentralized characteristics enable users to access, sift through, and disseminate a wide range of information with different levels of quality in real time (Hosni, Li, & Ahmad, 2020). This free flow of information fosters an efficient communication environment. It also becomes a breeding ground for the rapid spread of

false information. Studies indicate that in OSNs, rumors, which are a particularly harmful form of information dissemination, experience exponential growth in both the speed of spread and the extent of influence (Fu, Song, & Feng, 2023). They not only cause the quick spread of cognitive biases within groups but might also result in a crisis of public trust (Afanasyeva, Bukrieva, Glyns' ka, Hlebova, & Falko, 2023), collective psychological panic, and even disruptions in the social system (Meng, Li, Huang, & Li, 2022; Wan, Wang, Pang, Wang, & Min, 2023a).

Existing research primarily follows two approaches in light of the threat online rumors pose to social stability. The first approach centers

* Corresponding author.

E-mail addresses: gqcq2010cqu@163.com (C. Gan), s230101008@stu.cqupt.edu.cn (H. Chen), zhuqy@cqupt.edu.cn (Q. Zhu), tangrscu@scu.edu.cn (R. Tang), dkj@dlut.edu.cn (D.K. Jain), fenglp@yeah.net (L. Feng), salvag@decsai.ugr.es (S. García).

<https://doi.org/10.1016/j.ijinfomgt.2025.102989>

Received 18 June 2025; Received in revised form 10 September 2025; Accepted 22 October 2025

Available online 3 November 2025

0268-4012/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

on the feature analysis of rumor content. It builds rumor detection models using techniques like data mining and artificial intelligence. This approach combines various detection technologies, including text analysis (Yan, Wang, & Zheng, 2023), image analysis (Wan, Wang, Pang, Wang, & Min, 2023b), video analysis (Cao, Tang, Lin, & Lin, 2025), and multimodal fusion detection methods (Yin, Chen, Zhu, & Jiang, 2025). By collecting real-time network data in the form of text, images, and videos and performing algorithmic analysis, it can directly recognize rumor information. Nonetheless, this method has three drawbacks. Firstly, it overly emphasizes the content features of rumors and overlooks the dynamic evolution during the dissemination process. Secondly, the detection is confined to the already-formed rumor content, making it hard to predict the potential spread trends of rumors in advance. Thirdly, it depends on pre-trained models and large-scale labeled data, leading to high costs for rumor detection and difficulties in conducting real-time analysis of rumor spread trends.

In response to the above limitations, the second research approach utilizes classical epidemiological models to establish a state transition framework (Xiao & Chen, 2001). Users are categorized into distinct states (Susceptible–Infective–Recovered) (Yu, Lu, Wang, & Li, 2021), and the rumor diffusion process is simulated via state transition probabilities (Guo & Yan, 2023; Tong, Jiang, Qiu, Luo, & Chen, 2023). While these models can illustrate the macroscopic propagation trend and capture the dynamic evolution process in real time, they have notable flaws. They assume that user decisions adhere to a linear state transition rule, overlooking the complex impact of multi-dimensional factors such as user emotional tendencies and behavioral cognition on the communication process. To address this shortcoming, subsequent studies have incorporated emotion modeling methods (Gao, Sun, Zhang, Liu, & Gao, 2023; Liu, 2024). By building the coupling relationship between emotional intensity and rumor propagation, these studies have confirmed the significant regulatory effect of emotional polarization on propagation dynamics. However, existing models typically adopt a single immune state assumption; once a user becomes immune, they will never spread rumors again. This does not align with real-world propagation scenarios. When users first encounter a rumor, they might enter a wait-and-see state and display temporary neutral behavior (temporary immunity). As online rumors evolve, the attitudes of these temporarily neutral and immune users toward the rumors may shift. They may either become forwarding users or permanent immune users. This neglect of the temporary immune state results in low prediction accuracy of the models.

This paper proposes a novel E-SF2I (Emotion-Susceptible-Forwarding-Temporary Immunity-Permanent Immunity) rumor propagation dynamics model to address the collaborative impact of users' emotional differentiation and temporary neutral behavior. This model breaks through the limitation of single-factor driving in traditional models and realizes the modeling of the rumor propagation process under the collaborative action of multiple factors. In addition, unlike previous studies, this paper conducts multi-level verification experiments to demonstrate the effectiveness of the proposed model. First, the theoretical results derived from the model are verified with the help of the real network structure datasets from the Stanford Network Analysis Platform (SNAP), such as Facebook, Twitter, and P2P networks. Then, on the real network dataset Twitter, 13 real rumor events are used to validate the effectiveness of the proposed model. Finally, for 3 representative real rumor events, the proposed model is compared with traditional and new models.

The main contributions of this paper can be summarized as follows:

- (1) The proposed E-SF2I model integrates the emotion-driven forwarding mechanism with dual immunity state transitions (temporary/permanent immunity), overcoming the quantitative limitations of traditional models in capturing the dynamic synergy between users' emotional differentiation and neutral behaviors.
- (2) The proposed study constructs a differential dynamic system with time-varying characteristics, demonstrates the existence and stability of the equilibrium, and establishes a theoretical closed-loop validation to support the model effectiveness.
- (3) Developed a rumor propagation simulation algorithm based on the E-SF2I framework and constructed a multi-level verification system. Leveraging real network datasets and 13 real rumor events, the study achieved integrated validation through theoretical derivation, computational simulation, and empirical research.

The subsequent parts of this paper are as follows. Section 2 introduces the related work. Section 3 introduces scenarios and constructs the dynamical model. Section 4 analyzes the proposed model theoretically. Section 5 verifies the theoretical results. Section 6 summarizes the work.

2. Related work

In recent years, the academic community has paid extensive attention to the problem of rumor propagation in social networks, and remarkable progress has been made in relevant research (Chen, Ma, & Yang, 2023; Wang et al., 2025). To detect online rumors effectively, scholars have put forward a variety of innovative approaches. In text rumor detection, Zheng, Dou, and Yan (2024) developed a rumor detection framework using hierarchical prototype contrastive learning (HPCL). This framework applies contrastive learning to create a set of dynamically-updatable hierarchical prototypes, which helps capture the hierarchical semantic structures of rumors. Regarding image detection, Yang et al. (2024) proposed a new dual-encoder network (D-Net) for image splicing forgery detection. The network uses unfixed and fixed encoders to differentiate between tampered and untampered areas in images, enabling the learning and detection of the structural features of forged information. Cao et al. (2025) concentrated on video detection. They built a Chinese short-video rumor dataset and proposed a new short-video rumor detection method based on causal graphs, which greatly improved the interpretability of short-video rumor detection results. In multimodal rumor detection, Sun et al. (2025) introduced a new dynamic graph-aware network. It integrates source, image, and dynamic dialogue graphs into a unified framework. Through multi-level dynamic interaction with evolving positions, it realizes multimodal rumor detection. These detection-based methods are good at capturing rumors and have targeted detection abilities. Nevertheless, they have high computational complexity and lack real-time feedback on the propagation process. As a result, it is difficult to adjust detection strategies promptly according to the dynamic changes in rumor spread, which may cause some delays in practical applications.

Considering the constraints of detection-based approaches, analyzing rumor propagation from the perspective of transmission dynamics has emerged as a novel research avenue. Yu, Wu, Zhang, Cong, and Wu (2024) introduced an innovative IRLTS (Ignorant-Rumor-Spreader-Lurker-Truth-Spreader-Stifler) rumor propagation model. This framework comprehensively accounts for latent psychological influences and truth-related factors embedded in rumors, enabling the prediction of rumor propagation trends and offering a robust theoretical foundation for devising strategies to manage such propagation. In another study, Yu et al. (2025) examined the varying psychological states of individuals during rumor dissemination, categorizing propagators into normal and malicious groups. They proposed the IS2TR model and demonstrated its efficacy in forecasting rumor propagation patterns. Sun, Wang, Chai, and Liu (2024) developed an SEIHR model that incorporates three distinct attitudes, integrating mechanisms such as forgetting, spontaneous recall, and stimulated recall. The accuracy of this model was validated through two diverse scenarios. These methodologies simulate rumor propagation by establishing models, segmenting populations into various states, and introducing parameters to describe the process. Nevertheless, in real-world rumor propagation, users frequently

Table 1
Comparison of related work.

References	Methods	Advances	Drawbacks
Cao et al. (2025), Sun et al. (2025), Yang et al. (2024), Zheng et al. (2024)	Rumor detection based on deep learning.	Strong ability to capture rumors, detection methods are highly targeted.	High computational complexity and lack of real-time feedback make it difficult to adjust the rumor detection strategy in time.
Sun et al. (2024), Yu et al. (2024, 2025)	Rumor propagation modeling method based on propagation dynamics.	The effects of different intervention strategies can be predicted by simulating the propagation process with differential equations of the propagation dynamics system.	The influence of user emotion on forwarding behavior is not considered, and emotion may accelerate the spread of rumors or change the transmission path.
Liqing and Shuqi (2021), Shen et al. (2022), Yin et al. (2021)	Modeling methods that consider user emotions.	Breaking through the limitations of traditional models that only focus on information dissemination, psychological mechanisms such as emotional resonance and polarization effect are introduced to get closer to user behavior in social media.	Existing models assume that users have only “forward” or “immune” options, ignoring the reality that a large number of users may temporarily maintain a neutral attitude and subsequently change their decision due to emotional fluctuations.

generate emotions and express them via comments or other means while forwarding rumors. In this context, rumors are intertwined with emotional elements and spread concurrently with their content.

Concerning the emotional fluctuations that occur during rumor propagation, Liqing and Shuqi (2021) introduced the SVIR rumor model by incorporating a vigilant state into the traditional SIR framework. This model emphasizes how individuals with heightened emotions can more effectively induce emotional resonance among users, thereby establishing an emotion-driven transition pathway from the infected (I) state to the recovered (R) state. Additionally, Yin et al. (2021) developed an emotion-oriented susceptible-forwarding-immune (E-SFI) propagation dynamics model based on the SFI structure. In this model, user emotions are intricately categorized into three distinct types — positive, neutral, and negative — and the influence of these emotional shifts on user behavior throughout the dissemination process is thoroughly examined. Furthermore, Shen, Tu, Guo, and Chen (2022) presented a dual-layer emotion-based susceptible-infected-recovered network (2E-SIR), and its practicality and efficacy were validated through experimental studies. These models have successfully integrated emotional differentiation factors into rumor propagation modeling. Compared with traditional models that only focus on information dissemination laws, they can more closely reflect the complexity of rumor dissemination in reality and provide new perspectives and methodological support for related research. However, these models still have certain limitations. Specifically, they assume that after users come into contact with rumors and the accompanying emotional information, they only have two choices: forwarding or permanent immunity. They ignore the third common behavior that users may take in actual scenarios—temporarily maintaining a neutral attitude without immediately forwarding rumors. It is worth noting that as time progresses and rumor dissemination deepens, these neutral users may experience emotional fluctuations due to external stimuli or other factors, thereby adjusting their decisions and participating in the further dissemination of rumors.

Through in-depth analysis and discussion of the aforementioned literature, in order to more clearly present the similarities, differences, and connections between various studies, the core content is systematically summarized in Table 1. In this paper, we propose an E-SF2I rumor propagation dynamics model that couples users' emotional differentiation with a dual immunity state. This model can not only describe the rumor propagation process but also depict the emotional evolution process of the user groups exposed to the rumor. Through dynamic analysis and multi-level verification, the validity of the model is demonstrated.

3. Model descriptions

3.1. Background

OSNs are a social structure in information networks, consisting of a collection of social individuals and complex and diverse connections

between individuals. Its uniqueness lies in endowing each user with the powerful ability to receive and forward information in real-time. Users can choose to read and understand the information they receive, whether it is shared by friends, official push notifications, or forwarded by strangers, and then repost it again to expand the scope of information dissemination. This ability not only promotes the rapid flow and sharing of information but also opens up convenient channels for the spread of rumors.

In OSNs, the process of rumor propagation is shown in Fig. 1. After rumors spread in OSNs, Different from general information, the emotions contained in online rumors are important determinants of propagation dynamics (Pröllochs, Bär, & Feuerriegel, 2021). Users may obtain rumor information through social networks or forward rumors influenced by their own emotions. The forwarded rumors will carry personal emotions, which will be reflected in comments and reposted captions during the dissemination process (Cai et al., 2019; Jia & Li, 2020). At the same time, some people are aware of the rumors but do not forward them, remaining in a state of lurking (Yu et al., 2024), these individuals remain temporarily neutral and will not spread the rumors, but may subsequently forward them with emotional content. Eventually, the rumors will be clarified, and users will gradually become immune to them and stop spreading them (Pan, Yan, Hu, He, & Wu, 2023). It can be found that when confronting rumors, users typically exhibit the following behavioral responses: (1) Believe the rumor and develop positive emotions, then repost it with positive sentiment; (2) Believe the rumor and generate negative emotions, subsequently reposting it with negative sentiment; (3) Initially remain neutral without immediate belief or reposting (temporary immunity), but may resume believing the rumor due to continuous propagation, regenerate emotions, and repost emotionally, or become permanently immune; Unlike susceptible users, temporarily immune users may have heard about the rumor, but they are hesitant due to uncertainty. Susceptible users are not yet aware of the rumor but are more sensitive; (4) If users learn the truth about the rumor, they will neither repost nor be affected by it, maintaining permanent immunity. Therefore, inspired by the work (Pröllochs et al., 2021; Yin et al., 2021; Yu et al., 2024), users usually exhibit the following five states in response to rumors.

- **Susceptible state (S):** Users are not aware of the rumors, but are sensitive to them.
- **Positive emotion forwarding state (F_p):** Users in this state will forward rumors with positive emotions.
- **Negative emotion forwarding state (F_n):** Users in this state will forward rumors with negative emotions.
- **Temporary immunity state (I_T):** The user maintains a neutral attitude and will not forward the rumors for the moment.
- **Permanent immune state (I_p):** As rumors spread, users become aware of their authenticity and no longer discuss or forward them.

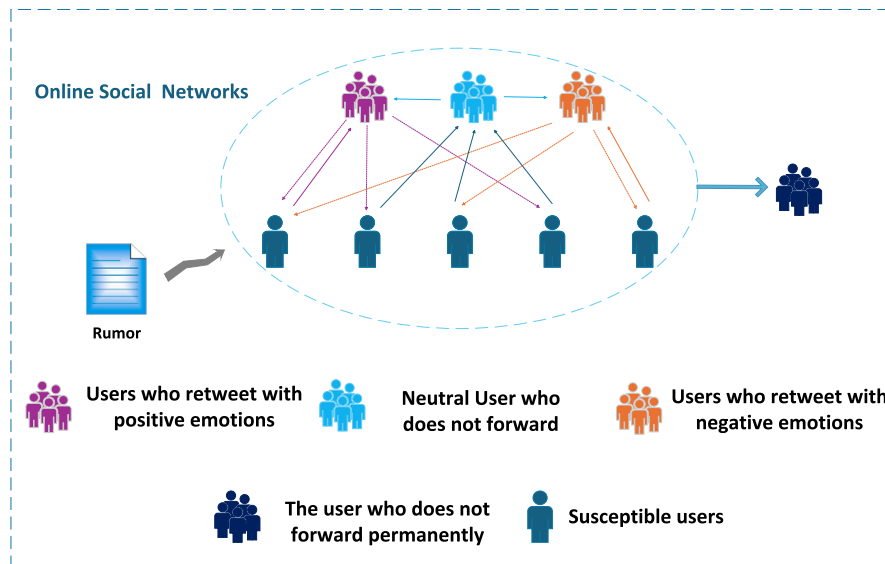


Fig. 1. Rumor propagation process in OSNs.

3.2. Motivations

Based on the aforementioned background, it is apparent that in the contemporary OSNs environment, the dissemination of rumors has emerged as a pressing issue requiring immediate attention. The propagation of rumors within OSNs constitutes a complex and dynamic process characterized by intricate interactions between emotional elements and network structures. When users encounter rumors, if they experience emotional resonance, they tend to disseminate both emotions and information via forwarding behavior. Some users may temporarily maintain a neutral attitude, questioning the veracity of the rumors (Evolving into temporary immunity); however, over time, they might lose this immunity and actively participate in rumor propagation, or they could transition directly into a permanently immune state. Users who discern the falsity of the rumors will attain permanent immunity and cease spreading them altogether. Concurrently, the network itself remains in a state of flux, with users either disconnecting from or joining the network. Over time, the entire rumor propagation process gradually reaches a stable equilibrium.

Given that users' forwarding behaviors are strongly driven by emotions — positive emotions accelerate propagation due to recognition, while negative emotions expand the scope of propagation due to anger, with differences in intensity, speed, and affected groups between the two — the model conforms to users' real forwarding behaviors through emotional differentiation, avoiding the oversimplification of propagation details caused by a single forwarding state. Meanwhile, the temporary immunity state captures the dynamic process of users' non-immediate decision-making, compensating for the prediction deviation of traditional models that overlook neutral, delayed behaviors, and better reflects the complexity of users' behaviors. The traditional SIR model has obvious limitations: it assumes that individuals entering the recovered state (R) permanently lose the ability to spread and fails to consider the diversity of user behaviors, thus being unable to reflect the situation where users first adopt a wait-and-see attitude after encountering rumors and then re-participate in propagation due to emotional arousal, etc. Additionally, it treats spreaders as a homogeneous group, ignoring the key impact of emotional motivations on propagation dynamics. However, in online social networks, users' emotional tendencies when forwarding rumors significantly affect the speed and scope of propagation. The motivation for these improvements is to address the mismatch between the simplified handling of traditional models and the complexity of user behaviors in online social

networks, thereby enhancing the accuracy of the model in predicting rumor propagation trends and its practical application value.

In light of these considerations, we employ the modeling framework of users' emotional differentiation coupled with dual immunity state mechanisms to construct a comprehensive dynamic model. This model not only elucidates the intricate dynamics of rumor propagation but also enables accurate prediction of its trends. Our objective is to identify the stable equilibrium of the rumor propagation process, develop predictive methodologies for the described scenarios, and implement preemptive intervention strategies to effectively curb the spread of rumors, thereby fostering a healthy and stable network environment.

3.3. Basic terms

To facilitate the comprehension of the subsequent analysis of the constructed propagation dynamics model, we present the following definitions.

- (1) **Dynamical System:** It is employed to depict the dynamical process of the system evolution over time (Robinson, 2012). In this paper, the target model pertains to the differential dynamic system of user emotional changes during the rumor-forwarding process. It is capable of describing the specific circumstances of rumor propagation at any given moment for each user, along with the concomitant emotional fluctuations and evolutions.
- (2) **Equilibrium:** Also referred to as a fixed point, it holds paramount significance in analyzing the local and global behaviors of dynamical systems (Robinson, 2012). In this paper, through conducting an in-depth analysis of the equilibrium of the dynamical system, we can gain a more thorough understanding of the dynamic behavior of the dynamical model and offer a novel perspective and methodology for the research on rumor propagation driven by emotions.
- (3) **Stability:** It refers to the long-term behavior of the equilibrium of a dynamical system (Robinson, 2012). The stability analysis of the equilibrium of the system model encompasses two crucial aspects: local asymptotic stability and global asymptotic stability. This is of considerable significance for an in-depth understanding and predictive analysis of the rumor propagation process accompanied by emotions. By exploring the stability of the equilibrium, we can precisely grasp the dynamic correlation between emotions and rumor propagation. This not only contributes to revealing the intrinsic mechanism of rumor propagation but also provides a robust basis for formulating effective rumor control strategies.

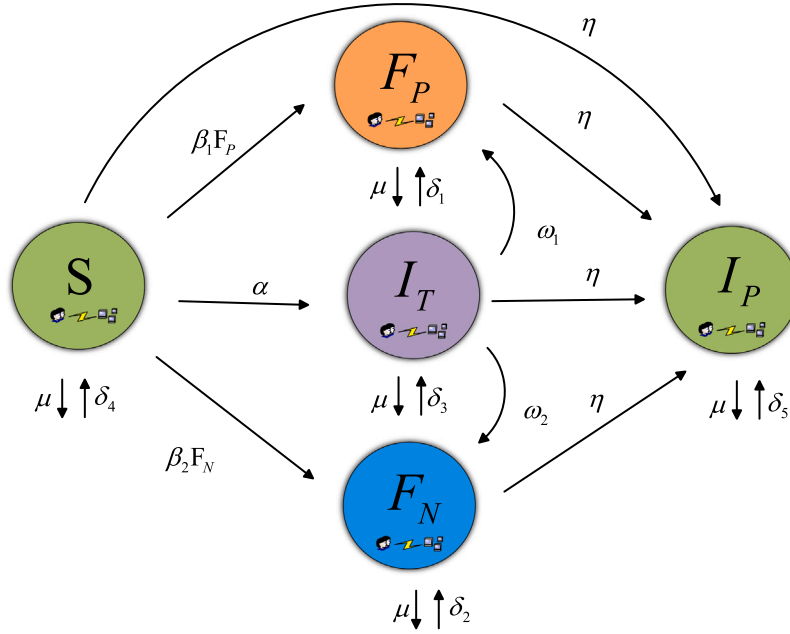


Fig. 2. State transfer diagram of rumor propagation process.

3.4. Basic assumptions

To describe the rumor propagation process in OSNs, based on the work (Guo & Yan, 2023; Pan et al., 2023; Yin et al., 2021; Yu et al., 2024), the following basic assumptions are made.

- (1) The probability of an online user going offline is μ , and the probabilities of an offline user rejoining the network are δ_1 , δ_2 , δ_3 , δ_4 , and δ_5 , respectively.
- (2) Under the influence of the forwarding users, the susceptible users will become forwarding users, among which the number of positive emotion forwarding users and negative emotion forwarding users will increase at the rates of $\beta_1 F_P$ and $\beta_2 F_N$, respectively.
- (3) Any susceptible user who receives a rumor may temporarily maintain a neutral attitude with a probability of α and not forward the rumor.
- (4) As the rumors spread and are discussed on the Internet, temporarily neutral users may regain their interest in the rumors, develop emotions, and forward them. They will respectively transform into users who forward rumors with positive emotions at a probability of ω_1 and users who forward rumors with negative emotions at a probability of ω_2 .
- (5) After the truth of the rumor was exposed, all users learned the truth of the rumor with a probability of η and would never forward the rumor again.

Based on the above basic assumptions, the state transitions can be obtained as shown in Fig. 2.

3.5. Model formulation

At time t , let $S(t)$, $F_P(t)$, $F_N(t)$, $I_T(t)$ and $I_P(t)$ respectively represent the number of users in S , F_P , F_N , I_T and I_P , and let $N(t)$ represent the total number of users. Then $N(t) = S(t) + F_P(t) + F_N(t) + I_T(t) + I_P(t)$. According to Fig. 2 and the basic assumptions, the rumor propagation

dynamics model can be described as:

$$\begin{cases} \frac{dS(t)}{dt} = \delta_4 - \mu S(t) - \alpha S(t) - \eta S(t) - \beta_1 S(t)F_P(t) - \beta_2 S(t)F_N(t), \\ \frac{dF_P(t)}{dt} = \delta_1 + \beta_1 S(t)F_P(t) + \omega_1 I_T(t) - \mu F_P(t) - \eta F_P(t), \\ \frac{dF_N(t)}{dt} = \delta_2 + \beta_2 S(t)F_N(t) + \omega_2 I_T(t) - \mu F_N(t) - \eta F_N(t), \\ \frac{dI_T(t)}{dt} = \delta_3 + \alpha S(t) - (\omega_1 + \omega_2 + \mu + \eta) I_T(t), \\ \frac{dI_P(t)}{dt} = \delta_5 + \eta (S(t) + F_P(t) + F_N(t) + I_T(t) + I_P(t)) - \mu I_P(t), \end{cases} \quad (1)$$

with initial condition $(S(0), F_P(0), F_N(0), I_T(0), I_P(0)) \in R_+^5$.

4. Model analysis

The previous section provided a detailed description of the rumor propagation dynamics model, and this section will further explore its dynamic characteristics, with a focus on analyzing the equilibrium and stability of the model. The proof is mainly based on the application of the dynamical system theory and the Lyapunov stability theory (Robinson, 2012).

4.1. System simplification

Given the complexity of the system (1), it is necessary to simplify it appropriately to ensure the smooth progress of subsequent analysis work. Note that $N(t) = S(t) + F_P(t) + F_N(t) + I_T(t) + I_P(t)$, system (1) can be rewritten as:

$$\begin{cases} \frac{dN(t)}{dt} = \delta - \frac{N(t)}{\mu}, \\ \frac{dS(t)}{dt} = \delta_4 - \mu S(t) - \alpha S(t) - \eta S(t) - \beta_1 S(t)F_P(t) - \beta_2 S(t)F_N(t), \\ \frac{dF_P(t)}{dt} = \delta_1 + \beta_1 S(t)F_P(t) + \omega_1 I_T(t) - \mu F_P(t) - \eta F_P(t), \\ \frac{dF_N(t)}{dt} = \delta_2 + \beta_2 S(t)F_N(t) + \omega_2 I_T(t) - \mu F_N(t) - \eta F_N(t), \\ \frac{dI_T(t)}{dt} = \delta_3 + \alpha S(t) - (\omega_1 + \omega_2 + \mu + \eta) I_T(t), \\ \frac{dI_P(t)}{dt} = \delta_5 + \eta (N(t) - I_P(t)) - \mu I_P(t), \end{cases} \quad (2)$$

where $\delta = \delta_1 + \delta_2 + \delta_3 + \delta_4 + \delta_5$ and the initial condition is $(N(0), S(0), F_P(0), F_N(0), I_T(0), I_P(0)) \in R_+^6$.

According to the first and sixth formulas in the system (2), $\lim_{t \rightarrow \infty} N = \frac{\delta}{\mu}$ and $\lim_{t \rightarrow \infty} I_P = \frac{\delta_5 + \mu N}{\eta}$ are obtained, according to the work (Robinson, 2012), we can simplify system (2) into a system with identical dynamical properties:

$$\begin{cases} \frac{dS(t)}{dt} = \delta_4 - \mu S(t) - \alpha S(t) - \eta S(t) - \beta_1 S(t) F_P(t) - \beta_2 S(t) F_N(t), \\ \frac{dF_P(t)}{dt} = \delta_1 + \beta_1 S(t) F_P(t) + \omega_1 I_T(t) - \mu F_P(t) - \eta F_P(t), \\ \frac{dF_N(t)}{dt} = \delta_2 + \beta_2 S(t) F_N(t) + \omega_2 I_T(t) - \mu F_N(t) - \eta F_N(t), \\ \frac{dI_T(t)}{dt} = \delta_3 + \alpha S(t) - (\omega_1 + \omega_2 + \mu + \eta) I_T(t), \end{cases} \quad (3)$$

where $N^* = \frac{\delta}{\mu}$, $I_P^* = \frac{\delta_5 + \mu N}{\eta}$ and the initial conditions are $(S(0), F_P(0), F_N(0), I_T(0)) \in R_+^4$. The positive invariant interval of the system (3) is:

$$\Omega = \{ (S(t), F_P(t), F_N(t), I_T(t)) \mid S(t) \geq 0, F_P(t) \geq 0, F_N(t) \geq 0, I_T(t) \geq 0, 0 \leq S(t) + F_P(t) + F_N(t) + I_T(t) \leq N^* \}. \quad (4)$$

Next, we only need to focus on the system (3) for dynamic analysis.

4.2. Equilibrium

This subsection will discuss the existence and uniqueness of the equilibrium of the system (3).

Theorem 1. System (3) has a unique equilibrium $E^* = (S^*, F_P^*, F_N^*, I_T^*)$.

Proof. Assume $E^* = (S^*, F_P^*, F_N^*, I_T^*)$ is an equilibrium of system (3), and according to the definition of the equilibrium, substitute $E^* = (S^*, F_P^*, F_N^*, I_T^*)$ into the system (3), we can get

$$\begin{cases} \delta_4 - \mu S^* - \alpha S^* - \eta S^* - \beta_1 S^* F_P^* - \beta_2 S^* F_N^* = 0, \\ \delta_1 + \beta_1 S^* F_P^* + \omega_1 I_T^* - \mu F_P^* - \eta F_P^* = 0, \\ \delta_2 + \beta_2 S^* F_N^* + \omega_2 I_T^* - \mu F_N^* - \eta F_N^* = 0, \\ \delta_3 + \alpha S^* - (\omega_1 + \omega_2 + \mu + \eta) I_T^* = 0. \end{cases} \quad (5)$$

By solving the second, third, and fourth equations of the system (5), one can obtain $I_T^* = AS^* + B$, $F_P^* = \frac{\omega_1 A}{-\beta_1} + \frac{C}{\mu + \eta - \beta_1 S^*}$ and $F_N^* = \frac{\omega_2 A}{-\beta_2} + \frac{D}{\mu + \eta - \beta_2 S^*}$, where $A = \frac{\delta_4}{\omega_1 + \omega_2 + \mu + \eta}$, $B = \frac{\delta_1}{\omega_1 + \omega_2 + \mu + \eta}$, $C = \omega_1 B + \frac{\omega_1 A(\mu + \eta)}{\beta_1}$ and $D = \omega_2 B + \frac{\omega_2 A(\mu + \eta)}{\beta_2}$. Then, substitute I_T^* , F_P^* and F_N^* into the first equation of system (5) to get

$$\delta_4 - \mu S^* - \alpha S^* - \eta S^* - \beta_1 S^* \left(\frac{\omega_1 A}{-\beta_1} + \frac{C}{\mu + \eta - \beta_1 S^*} \right) - \beta_2 S^* \left(\frac{\omega_2 A}{-\beta_2} + \frac{D}{\mu + \eta - \beta_2 S^*} \right) = 0. \quad (6)$$

Let $x = S^*$, then one can find the expression of $F(x)$ for x :

$$F(x) = \delta_4 + C + D - \mu x - \alpha x - \eta x + (\omega_1 + \omega_2) A x - \frac{C(\mu + \eta)}{\mu + \eta - \beta_1 x} - \frac{D(\mu + \eta)}{\mu + \eta - \beta_2 x}, \quad (7)$$

where $F(0) = \delta_4 > 0$. Then, take the derivative of $F(x)$, then

$$F(x)' = -\mu - \alpha - \eta + (\omega_1 + \omega_2) A - \frac{\beta_1 C(\mu + \eta)}{(\mu + \eta - \beta_1 x)^2} - \frac{\beta_2 D(\mu + \eta)}{(\mu + \eta - \beta_2 x)^2}. \quad (8)$$

Noting that $-\mu - \alpha - \eta + (\omega_1 + \omega_2) A < -\mu - \alpha - \eta + \alpha$, so $F(x)' < 0$ and $F(x)$ monotonically decreases at $x > 0$. According to $F(0) > 0$, $F(x)$ has a unique zero at $x > 0$. Then one can conclude that there is a unique $S^* > 0$ for Eq. (6) to be true. And according to Eq. (4), it can be seen that $S(t) > 0$ is always true, so there is a unique equilibrium $E^* = (S^*, F_P^*, F_N^*, I_T^*)$ in the system (3).

4.3. Local stability analysis of the equilibrium

Theorem 1 states that the system (3) has a unique equilibrium, and this subsection will discuss the local stability of the equilibrium.

Theorem 2. The equilibrium E^* of the system (3) is locally asymptotically stable.

Proof. The Jacobi matrix of system (3) in E^* is

$$J_{E^*} = \begin{pmatrix} J_1 & -\beta_1 S^* & -\beta_2 S^* & 0 \\ \beta_1 F_P^* & J_2 & 0 & \omega_1 \\ \beta_2 F_N^* & 0 & J_3 & \omega_2 \\ \alpha & 0 & 0 & J_4 \end{pmatrix}, \quad (9)$$

where

$$\begin{cases} J_1 = -\mu - \alpha - \eta - \beta_1 F_P^* - \beta_2 F_N^* < 0, \\ J_2 = -\mu - \eta + \beta_1 S^* < 0, \\ J_3 = -\mu - \eta + \beta_2 S^* < 0, \\ J_4 = -\mu - \eta - \omega_1 - \omega_2 < 0. \end{cases} \quad (10)$$

According to the sufficient and necessary conditions of the negative definite matrix, the leading Principle Submatrix $D_i(-1)^k > 0$ of the matrix, that is, the leading Principle Submatrix $D_i < 0$ of odd order and the leading Principle Submatrix $D_i > 0$ of even order.

One can calculate the leading Principle Submatrix of Eq. (9) to get: Order 1 leading Principle Submatrix

$$D_1 = J_1 < 0. \quad (11)$$

Order 2 leading Principle Submatrix

$$D_2 = \begin{pmatrix} J_1 & -\beta_1 S^* \\ \beta_1 F_P^* & J_2 \end{pmatrix} > 0. \quad (12)$$

Order 3 leading Principle Submatrix

$$D_3 = \begin{pmatrix} J_1 & -\beta_1 S^* & -\beta_2 S^* \\ \beta_1 F_P^* & J_2 & 0 \\ \beta_2 F_N^* & 0 & J_3 \end{pmatrix} < 0. \quad (13)$$

Order 4 leading Principle Submatrix

$$D_4 = \begin{pmatrix} J_1 & -\beta_1 S^* & -\beta_2 S^* & 0 \\ \beta_1 F_P^* & J_2 & 0 & \omega_1 \\ \beta_2 F_N^* & 0 & J_3 & \omega_2 \\ \alpha & 0 & 0 & J_4 \end{pmatrix} > 0. \quad (14)$$

According to Eqs. (11)–(14), Eq. (9) is a negative definite matrix, and all roots of the characteristic equation have negative real parts. Therefore, the equilibrium of the system (3) is locally asymptotically stable, and the proposition is proved.

4.4. Global stability analysis of the equilibrium

Theorem 2 verifies the local asymptotic stability of the equilibrium point, and we will now discuss the global stability of the equilibrium. The system (3) in the feasible domain R_+^4 has an invariant manifold

$$\Gamma = (S, F_P, F_N, I_T) \in R_+^4 : S + F_P + F_N + I_T = C, \quad (15)$$

where $C = N^* - I_P^*$. The invariance of Γ is determined by the following formula

$$(S + F_P + F_N + I_T - C)' = (-\mu - \eta)(S + F_P + F_N + I_T - C). \quad (16)$$

According to Eq. (16), $g(x) = S + F_P + F_N + I_T - C$, $N(x) = -\mu - \eta$, where $(S, F_P, F_N, I_T) \in R_+^4$, satisfies Ω , and $m = rk\left(\frac{\partial g}{\partial x}\right) = 1$. Let $\bar{\Gamma}$ and $\hat{\Gamma}$ be the closure and interior of the invariant manifold Γ in the hyperplane $S + F_P + F_N + I_T = C$, the following Lemma can be obtained.

Lemma 1 (Li & Muldowney, 2000). The system has a unique equilibrium $E^* = (S^*, F_P^*, F_N^*, I_T^*) \in \hat{\Gamma}$, and E^* is locally asymptotically stable. If $\tilde{\mu}(t)$ exists, make $\lim_{T \rightarrow \infty} \int_0^T \tilde{\mu}(\tau) - v(\tau) d\tau < 0$ true, the unique equilibrium E^* of the system is globally asymptotically stable.

Theorem 3. The unique equilibrium E^* is globally asymptotically stable.

Proof. Let $f(x)$ represent the vector field defined by the system (3), and $x = (S, F_P, F_N, I_T)$, according to Lemma 1, has

$$\dot{x} = f(x), \quad \dot{z} = \frac{\partial f^{[3]}}{\partial x}(x)z, \quad (17)$$

where $z = (z_1, z_2, z_3, z_4) \in \mathbb{R}^4 \cong \mathbb{R}_+^4$.

According to the third composite matrix calculation method, $\frac{\partial f^{[3]}}{\partial x}$ of the system (3) is

$$\begin{pmatrix} m_1 + m_2 + m_3 & \omega_2 & -\omega_1 & 0 \\ 0 & m_1 + m_2 + m_4 & 0 & \beta_2 S \\ 0 & 0 & m_1 + m_3 + m_4 & -\beta_1 S \\ \alpha & -\beta_2 F_N & \beta_1 F_P & m_2 + m_3 + m_4 \end{pmatrix}, \quad (18)$$

where $m_1 = -\mu - \alpha - \eta - \beta_1 F_P - \beta_2 F_N < 0$, $m_2 = -\mu - \eta + \beta_1 S < 0$, $m_3 = -\mu - \eta + \beta_2 S < 0$, $m_4 = -\mu - \eta - \omega_1 - \omega_2 < 0$.

Next, we set

$$V(x, z) = \max \left\{ b_1 |z_1| + |z_2|, \frac{S}{F_N} (b_2 |z_3| + |z_4|) \right\}, \quad (19)$$

where b_1 and b_2 satisfy $\frac{\alpha}{\omega_1 + \omega_2} \leq b_1 \leq \frac{\omega_1 + \omega_2 + \mu + \eta}{\omega_2}$ and $\frac{\beta_1 F_P}{\mu + \eta + \alpha + \beta_1 F_P + \beta_2 F_N} < 1 \leq b_2$. $V(x, z) = V(x(t), z(t))$ where $(x(t), z(t))$ is the solution to Eq. (17), at any time t exists $\dot{V}(x, z) = V'(x(t), z(t))$. Then the differential inequality is obtained according to Eq. (17) and (18) as follows.

$$\begin{cases} D_+ b_1 |z_1(t)| \leq (m_1 + m_2 + m_3) b_1 |z_1(t)| + \omega_2 b_1 |z_2(t)| - \omega_1 b_1 |z_3(t)|, \\ D_+ |z_2(t)| \leq (m_1 + m_2 + m_4) |z_2(t)| + \beta_2 F_N(t) \frac{S(t)}{F_N(t)} |z_4(t)|, \\ D_+ b_2 |z_3(t)| \leq (m_1 + m_3 + m_4) b_2 |z_3(t)| - \beta_1 b_2 S(t) |z_4(t)|, \\ D_+ |z_4(t)| \leq \frac{\alpha}{b_1} b_1 |z_1(t)| - \beta_2 F_N(t) |z_2(t)| \\ \quad + \frac{\beta_1 F_P(t)}{b_2} b_2 |z_3(t)| + (m_2 + m_3 + m_4) |z_4(t)|. \end{cases} \quad (20)$$

According to Eq. (20), set $v_1(t) = b_1 |z_1(t)| + |z_2(t)|$, $v_2(t) = \frac{S(t)}{F_N(t)} (b_2 |z_3(t)| + |z_4(t)|)$, obtain $D_+ |v_1(t)|$ and $D_+ |v_2(t)|$. Since m_2 and m_3 cannot determine the range of values, the following two situations exist.

Case 1: at any time t , $m_2 < 0$ and $m_3 < 0$, by Eq. (20), one can get

$$\begin{cases} D_+ v_1(t) \leq (m_1 + m_2) v_1(t) + \beta_2 F_N(t) v_2(t), \\ D_+ v_2(t) \leq \left(\frac{S'(t)}{S(t)} - \frac{F'_N(t)}{F_N(t)} + m_3 + m_4 \right) v_2(t) + \frac{\alpha S(t)}{b_1 F_N(t)} v_1(t), \end{cases} \quad (21)$$

so $D_+ V(t) \leq \tilde{\mu}(t) V(t)$, where $\tilde{\mu}(t) = \max \{g_1(t), g_2(t)\}$, and $g_1(t) = m_1 + m_2 + \beta_2 F_N(t)$, $g_2(t) = \frac{S'(t)}{S(t)} - \frac{F'_N(t)}{F_N(t)} + m_3 + m_4 + \frac{\alpha S(t)}{b_1 F_N(t)}$.

From system (3) one can get $\frac{S(t)}{F_N(t)} = \frac{F'_N(t)}{\beta_2 F_N^2(t)} + \frac{\mu + \eta}{\beta_2 F_N(t)} - \frac{\delta_1 + \omega_2 I_T(t)}{\beta_2 F_N^2(t)} < \frac{F'_N(t)}{\beta_2 F_N^2(t)} + \frac{\mu + \eta}{\beta_2 F_N(t)}$, and because of $v(t) = -\mu - \eta$, so

$$\lim_{T \rightarrow \infty} \int_0^t \tilde{\mu}(\tau) - v(\tau) d\tau = \lim_{T \rightarrow \infty} \int_0^t \max \{g_1(\tau), g_2(\tau)\} + \mu + \eta d\tau.$$

If $g_1(t)$ is larger, $\lim_{T \rightarrow \infty} \int_0^t \tilde{\mu}(\tau) - v(\tau) d\tau = \lim_{T \rightarrow \infty} \int_0^t \beta_1 S(\tau) - \mu - \eta - \beta_1 F_P(\tau) - \alpha d\tau < 0$. If $g_2(t)$ is larger, $\lim_{T \rightarrow \infty} \int_0^t \tilde{\mu}(\tau) - v(\tau) d\tau = \lim_{T \rightarrow \infty} \int_0^t \beta_2 S(\tau) - \mu - \eta - \omega_1 - \omega_2 - (1 - \frac{\alpha}{b_1}) \frac{F'_N(\tau)}{F_N^2(\tau)} + \frac{\mu + \eta}{F_N(\tau)} \frac{\alpha}{b_1} + \frac{S'(\tau)}{S(\tau)} d\tau < 0$.

So at any time t , $m_2 < 0$ and $m_3 < 0$, there is $\lim_{T \rightarrow \infty} \int_0^t \tilde{\mu}(\tau) - v(\tau) d\tau < 0$.

Case 2: when any time t , $m_2 > 0$ and $m_3 > 0$ or some time t , $m_2 < 0$ and $m_3 < 0$, and the rest time $m_2 > 0$ and $m_3 > 0$, we consider

any time, $m_2 > 0$ and $m_3 > 0$ to be convenient, one can get

$$\begin{cases} D_+ v_1(t) \leq (m_1 + m_2 + m_3) v_1(t) + \beta_2 F_N(t) v_2(t), \\ D_+ v_2(t) \leq \left(\frac{S'(t)}{S(t)} - \frac{F'_N(t)}{F_N(t)} + m_2 + m_3 + m_4 \right) v_2(t) + \frac{\alpha S(t)}{b_1 F_N(t)} v_1(t), \end{cases} \quad (22)$$

so $D_+ V(t) \leq \tilde{\mu}(t) V(t)$, where $\tilde{\mu}(t) = \max \{g_1(t), g_2(t)\}$, and $g_1(t) = m_1 + m_2 + m_3 + \beta_2 F_N(t)$, $g_2(t) = \frac{S'(t)}{S(t)} - \frac{F'_N(t)}{F_N(t)} + m_2 + m_3 + m_4 + \frac{\alpha S(t)}{b_1 F_N(t)}$.

From system (3) one can get $\beta_1 S(t) = \frac{F'_P(t) - \delta_1 - \omega_1 I_T(t)}{F_P(t)} + \mu + \eta < \frac{F'_P(t)}{F_P(t)} + \mu + \eta$, $\beta_2 S(t) = \frac{F'_N(t) - \delta_2 - \omega_2 I_T(t)}{F_N(t)} + \mu + \eta < \frac{F'_N(t)}{F_N(t)} + \mu + \eta$.

If $g_1(t)$ is larger, $\lim_{T \rightarrow \infty} \int_0^t \tilde{\mu}(\tau) - v(\tau) d\tau = \lim_{T \rightarrow \infty} \int_0^t \frac{F'_P(\tau)}{F_P(\tau)} + \frac{F'_N(\tau)}{F_N(\tau)} - \beta_1 F_P(\tau) - \alpha d\tau < 0$. If $g_2(t)$ is larger, $\lim_{T \rightarrow \infty} \int_0^t \tilde{\mu}(\tau) - v(\tau) d\tau = \lim_{T \rightarrow \infty} \int_0^t \frac{F'_P(\tau)}{F_P(\tau)} - \omega_1 - \omega_2 + \frac{\alpha}{b_1} \left(\frac{F'_N(\tau)}{F_N^2(\tau)} + \frac{\mu + \eta}{F_N(\tau)} \right) + \frac{S'(\tau)}{S(\tau)} d\tau < 0$. So at any time t , $m_2 > 0$ and $m_3 > 0$, there is $\lim_{T \rightarrow \infty} \int_0^t \tilde{\mu}(\tau) - v(\tau) d\tau < 0$.

In summary, at any time t , there is $\lim_{T \rightarrow \infty} \int_0^t \tilde{\mu}(\tau) - v(\tau) d\tau < 0$, the unique equilibrium E^* of the system (3) is globally asymptotically stable.

Remark 1. Theorems 1–3 describe the propagation characteristics of rumors in OSNs, indicating that once rumors are spread in OSNs, they will persist for a considerable period, and any single rumor-refuting measure cannot eliminate their spread and accompanying effects. Therefore, attention should be shifted from eliminating rumors to trying to mitigate the negative impacts of rumor spreading. By observing the expressions of F_P and F_N in Theorem 1, one can find a potential way to optimize the control of rumor propagation, and some parameters in the expression have a significant impact on the rumor propagation process and its influence range. Specifically, by adjusting these key parameters, the proportion of users forwarding rumors can be reduced to a relatively low level, and the spread scope of rumors can be effectively controlled. This provides an important direction for the suppression of rumor propagation in OSNs and develops effective strategies and methods to minimize the negative impact of rumors.

5. Experiments

In this section, we use multiple real network datasets and real rumor events to conduct multi-level experimental verification of the model and verify the effectiveness of the proposed E-SF2I model, which can be applied to real networks and predict real rumor events and has performance advantages compared with other models. Firstly, we perform numerical simulations based on differential equations and design algorithms to perform a real network simulation of the model on real network datasets. Then, we fit the model parameters according to the real rumor events and verify the rumor propagation trend on the real network. Among them, the fitting method adopts the ideas from traditional machine learning, dividing the events into segments for fitting and verification in a certain proportion (Gan et al., 2025). Secondly, the proposed model fits and predicts the real rumor events with several experimental models and compares the degree of agreement with the real rumor events to verify the superiority of the proposed model. Finally, we analyze several model parameters and discuss their effects on rumor propagation.

Table 2

A summary of the characteristics of the datasets.

Dataset	Number of nodes	Number of edges
Facebook dataset	4039	88 234
Twitter dataset	81 306	1768149
Peer-to-Peer (P2P) dataset	62586	147 892

5.1. Experimental setup

All experiments are performed on the Python 3.9.0 platform. The Facebook dataset (Mcauley & Leskovec, 2012), Twitter dataset (Mcauley & Leskovec, 2012), and P2P dataset (Leskovec, Kleinberg, & Faloutsos, 2007; Ripeanu, Foster, & Iamnitchi, 2002) are adopted to simulate the real network to verify the stability of the proposed model. SNAP provided these three datasets (Bodaghi & Oliveira, 2020). Dataset nodes (users) and edges (connected networks) are described in Table 2. Facebook, as a highly clustered social network of close friends, has a relatively small and closed network scale. As a private social platform, its strong connection feature can effectively verify the dynamic characteristics of emotion-driven dissemination in a closely-knit community. Twitter, as an open and large-scale social media network, can facilitate the dissemination of public content and can be used to test whether the model can predict the real rumor events and estimate the accuracy of the dissemination scale in large-scale networks. P2P, this decentralized and non-social network, strips away the social relationship elements and specifically examines the model's universality of the emotional dissemination mechanism in a pure network. This multi-level verification strategy not only demonstrates the adaptability of the model in different network topologies, but also reveals the regulatory effects of social relationship density and network openness on rumor emotional dissemination by comparing the sensitivity differences of dissemination parameters among the three types of networks. It is used to prove that the propagation mechanism of emotional differentiation and dual immune state coupling is a fundamental dynamic characteristic in any network topology. Because the number of nodes and connected edges in the dataset is fixed, we set the probability parameter of the user leaving and accessing the network to 0, which can minimize the simulation error.

Algorithm 1: Evolutionary algorithm of rumor forwarding propagation dynamic model.

Input: Example Set initial network $G = (V, E)$ with the dataset.

Output: The number of nodes S, F_P, F_N, I_T, I_P .

- 1: The initial number of nodes in different states is randomly assigned.
- 2: Set the number of iterations to t .
- 3: **for** $t = 1$ to j , step size is 1 **do**
- 4: **if** node i is in S state at time t and does not leave the network at time $t+1$, **then**
- 5: traversing the neighbor nodes, i becomes F_P or F_N with the introduction of β_1 or β_2
under the influence of F_P or F_N , while node i becomes an I_T or I_P node with the
probability of α or η ;
- 6: **else** the status of node i remains unchanged;
- 7: **if** node i is in the F_P state at time t and does not leave the network at time $t+1$, **then**

- 8: node i becomes node I_P with probability η ;
- 9: **else** the status of node i remains unchanged;
- 10: **if** node i is in the F_N state at time t and does not leave the network at time $t+1$, **then**
- 11: node i becomes node I_P with probability η ;
- 12: **else** the status of node i remains unchanged;
- 13: **if** node i is in the I_T state at time t and does not leave the network at time $t+1$, **then**
- 14: node i becomes node F_P, F_N or I_P with probability of ω_1, ω_2 or η ;
- 15: **else** the status of node i remains unchanged;
- 16: **if** node i is in the I_P state at time t and does not leave the network at time $t+1$, **then**
- 17: the status of node i remains unchanged;
- 18: **end if**
- 19: **end**
- 20: Calculate the number of nodes S, F_P, F_N, I_T, I_P and return the result.

We design an algorithm based on real network datasets to achieve theoretical verification, as shown in Algorithm 1. Based on the E-SF21 model, this algorithm simulates the dynamic process of rumor propagation in OSNs. Through node state transfer and network dynamic connection mechanism, it quantifies the influence of user emotion differentiation (positive/negative) and dual immunity state coupling on rumor diffusion. Through multiple iterations, the algorithm dynamically updates node states and finally outputs the number of nodes in each state, indicating the number of users in the corresponding state. The evolutionary law of rumor propagation is revealed.

It is convenient to distinguish the experimental results of the two simulation methods. Let $S(t), F_P(t), F_N(t), I_T(t), I_P(t)$ represent the experimental results of differential equation simulation, and $S_e(t), F_{Pe}(t), F_{Ne}(t), I_{Te}(t), I_{Pe}(t)$ represent the experimental results of Algorithm 1 simulation in real networks.

For the real rumor dataset verification of the proposed model, we used a new rumor dataset on Twitter, NERT (Bodaghi, 2019), which contains 12 different rumor events. Any row in the dataset is about the tweet/retweet of the corresponding rumor event, and each column is about the relevant information of the corresponding tweet/retweet. A rumor dataset¹ with emotional labels was also used, which contained 1,048,575 rumor tweets from more than 2,000 events, among which 22,155 tweets with emotional labels. Since the number of tweets with emotional labels for a single event was small, we chose events r21 and r55 in the dataset for the experiment. According to the work (Yin et al., 2021), we used the same method to fit the parameters of the model with some data of each event, verify the model with the remaining data, and simulate the real network with the fitting parameters. Where fitting is to fit the system parameters through the error function, which is:

$$f(X) = \sum_{i=1}^N (X_{True,i} - X_i)^2. \quad (23)$$

5.2. Theoretical verification using real network datasets

Considering system (1) and Theorem 3, we will verify the stability of the proposed model in terms of both initial conditions and system parameters: (i) simulation with different initial conditions considering the same system parameters, and (ii) simulation with different system parameters considering the same initial conditions.

Scenario 1. Model stability verification based on the Facebook dataset network structure.

¹ https://www.dropbox.com/scl/fi/1btjbzghi9gheleg3yqki/FalseNews_Code_Data.zip?rlkey=wtf07mtybhnaeapd0tnlsuqx&e=1&dl=0

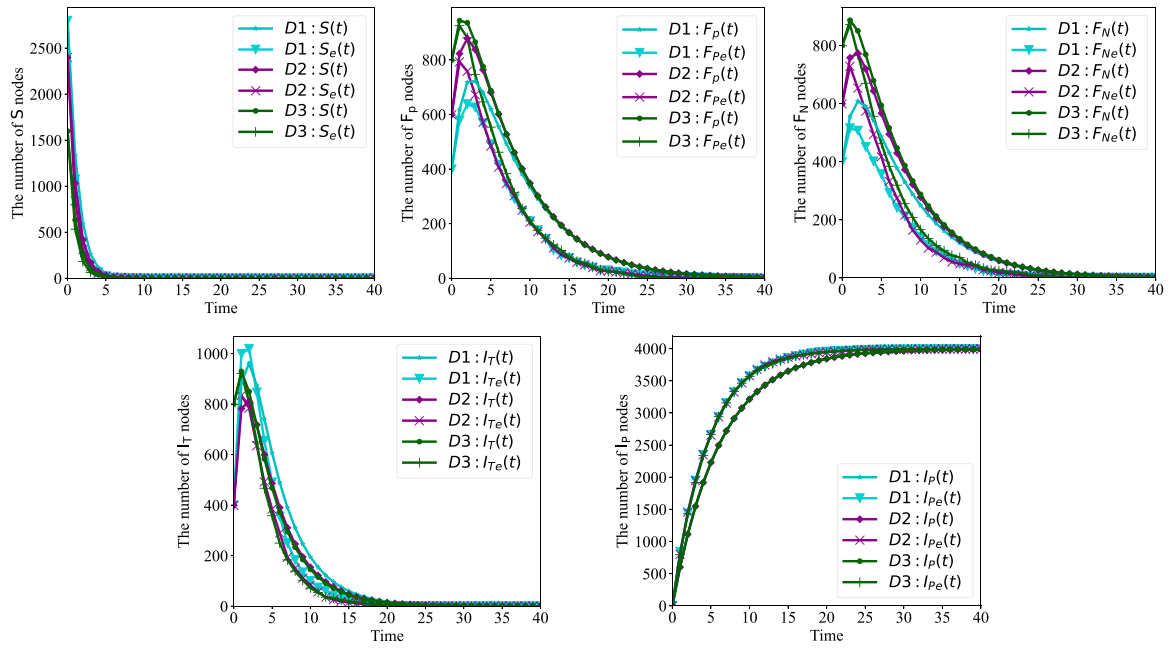


Fig. 3. Evolution of system (1) under given conditions in Experiment 1.

Table 3

System parameters in Experiment 1.

Parameter	β_1	β_2	ω_1	ω_2	α	η
Value	0.33×10^{-3}	0.28×10^{-3}	0.058	0.033	0.4	0.2

Table 4

System parameters in Experiment 2.

Parameter	β_1	β_2	ω_1	ω_2	α	η
Value1	0.33×10^{-3}	0.28×10^{-3}	0.058	0.033	0.4	0.2
Value2	0.40×10^{-3}	0.33×10^{-3}	0.030	0.049	0.35	0.25
Value3	0.36×10^{-3}	0.25×10^{-3}	0.044	0.058	0.45	0.3

Experiment 1: Consider system (1), use the same parameters as shown in Table 3, and change the initial conditions as follows:

- (D1) $(S(0), F_P(0), F_N(0), I_T(0), I_P(0)) = (2798, 399, 399, 399, 0)$,
 (D2) $(S(0), F_P(0), F_N(0), I_T(0), I_P(0)) = (2397, 599, 599, 399, 0)$,
 (D3) $(S(0), F_P(0), F_N(0), I_T(0), I_P(0)) = (1598, 799, 799, 799, 0)$.

Fig. 3 illustrates the evolution of the system (1) under the conditions specified in Experiment 1. The figure demonstrates that the numerical simulation results of the differential equation align closely with the real network simulation results from the Facebook dataset, validating the accuracy of the proposed rumor propagation model. Regardless of the variations in initial conditions, the state of each user eventually stabilizes, indicating that the system stability is unaffected by changes in initial conditions. Consequently, the proposed model effectively predicts rumor propagation behavior within real networks. Furthermore, stabilizing all user state values confirms the equilibrium existence, uniqueness, and stability, thereby satisfying Theorems 1–3.

Experiment 2: Consider system (1) and change the system parameters as shown in Table 4, using the same initial conditions: $(S(0), F_P(0), F_N(0), I_T(0), I_P(0)) = (2798, 399, 399, 399, 0)$.

Fig. 4 shows the evolution results of the system (1) under the given conditions of Experiment 2. We fixed the initial conditions and changed the system parameters to get different evolution results. However, the real network simulation of the Facebook dataset and the numerical simulation of the differential equation show similar evolution trends for each state, which indicates that changing the system parameters can

Table 5

System parameters in Experiment 3.

Parameter	β_1	β_2	ω_1	ω_2	α	η
Value	0.48×10^{-4}	0.53×10^{-4}	0.58×10^{-3}	0.29×10^{-3}	0.55	0.22

Table 6

System parameters in Experiment 4.

Parameter	β_1	β_2	ω_1	ω_2	α	η
Value1	0.48×10^{-4}	0.53×10^{-4}	0.58×10^{-3}	0.29×10^{-3}	0.55	0.22
Value2	0.60×10^{-4}	0.30×10^{-4}	0.30×10^{-3}	0.49×10^{-3}	0.45	0.27
Value3	0.36×10^{-4}	0.44×10^{-4}	0.44×10^{-3}	0.58×10^{-3}	0.48	0.3

reduce the spread scale of rumors. The rapid decline in the number of S users suggests that many uninformed people will be exposed to rumors in the early stage of rumor propagation.

Scenario 2. Model stability verification based on the Twitter dataset network structure.

Experiment 3: Consider system (1), use the same parameters as shown in Table 5, and change the initial conditions as follows:

- (D1) $(S(0), F_P(0), F_N(0), I_T(0), I_P(0)) = (54776, 7682, 7682, 7682, 0)$,
 (D2) $(S(0), F_P(0), F_N(0), I_T(0), I_P(0)) = (46094, 11523, 11523, 7682, 0)$,
 (D3) $(S(0), F_P(0), F_N(0), I_T(0), I_P(0)) = (34570, 15364, 15364, 11524, 0)$.

Fig. 5 presents the evolution results of the system (1) under specific conditions in Experiment 3. The numerical simulation results of the differential equation align closely with the real network simulation outcomes from the Twitter dataset. Similar findings from both Experiment 3 and Experiment 1 further confirm the applicability and reliability of the proposed model across different social networks.

Experiment 4: Consider system (1) and change the system parameters as shown in Table 6, using the same initial conditions: $(S(0), F_P(0), F_N(0), I_T(0), I_P(0)) = (54776, 7682, 7682, 7682, 0)$.

Fig. 6 depicts the evolution results of the system (1) under specific conditions in Experiment 4. It demonstrates that variations in system parameters significantly impact rumor propagation in the real Twitter network. The results align with the conclusions drawn in Experiment 2, further reinforcing the findings.

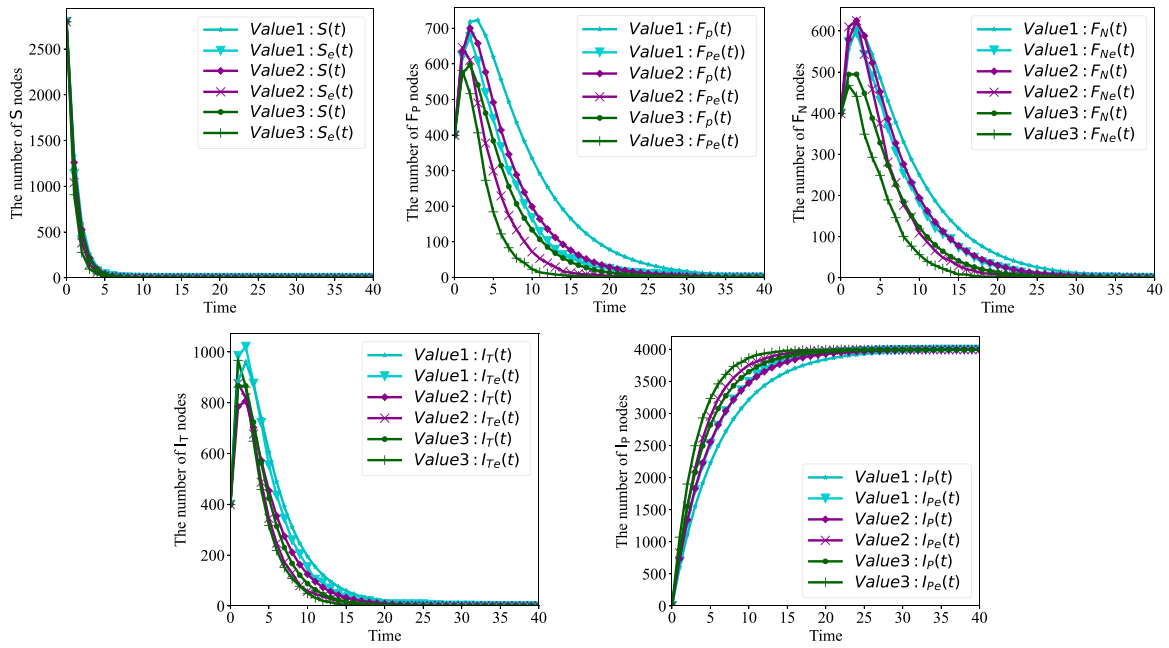


Fig. 4. Evolution of system (1) under given conditions in Experiment 2.

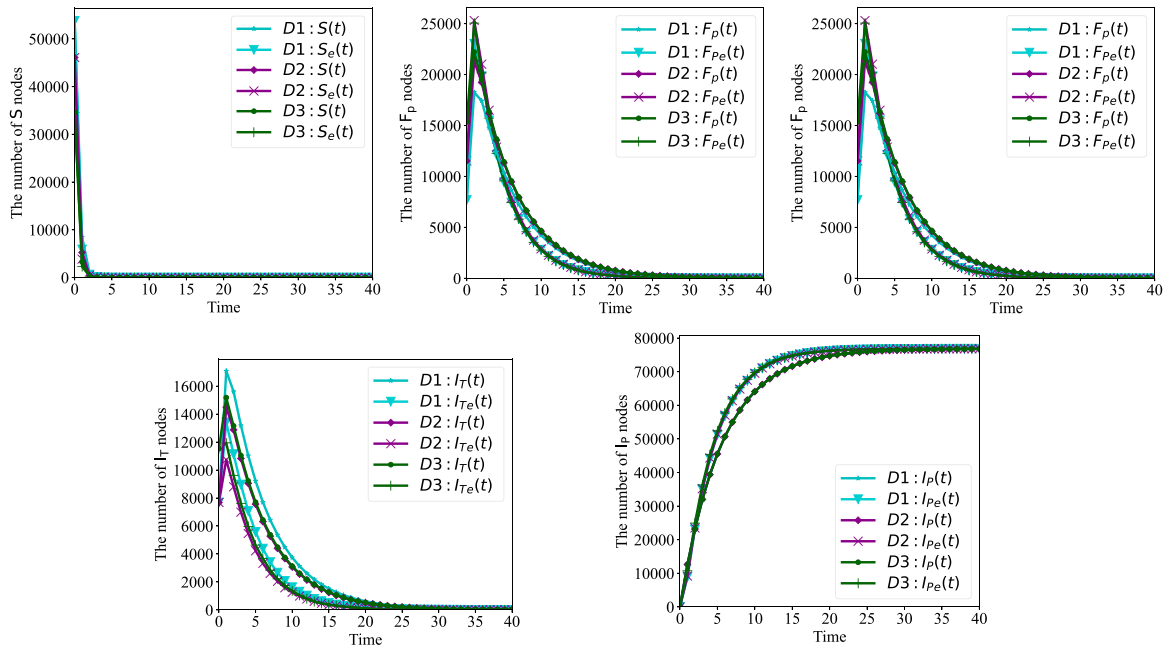


Fig. 5. Evolution of system (1) under given conditions in Experiment 3.

Scenario 3. Model stability verification based on P2P dataset network structure.

Experiment 5: Consider system (1), use the same parameters as shown in Table 7, and change the initial conditions as follows:

$$\begin{aligned} (D1) \quad & (S(0), F_p(0), F_N(0), I_T(0), I_P(0)) = (43812, 6258, 6258, 6258, 0), \\ (D2) \quad & (S(0), F_p(0), F_N(0), I_T(0), I_P(0)) = (37552, 9388, 9388, 6258, 0), \\ (D3) \quad & (S(0), F_p(0), F_N(0), I_T(0), I_P(0)) = (28164, 12517, 12517, 9388, 0). \end{aligned}$$

Fig. 7 illustrates the evolution results of the system (1) under specific conditions in Experiment 5. Consistent with the findings from Experiments 1 and 3, the number of corresponding users in the P2P dataset exhibits a similar trend. This strongly validates the practicality and applicability of the proposed model in real networks.

Table 7

System parameters in Experiment 5.

Parameter	β_1	β_2	ω_1	ω_2	α	η
Value	0.50×10^{-4}	0.80×10^{-4}	0.25×10^{-2}	0.4×10^{-3}	0.50	0.30

Experiment 6: Considering system (1), change the system parameters as shown in Table 8, using the same initial conditions: $(S(0), F_p(0), F_N(0), I_T(0), I_P(0)) = (43812, 6258, 6258, 6258, 0)$.

Fig. 8 presents the evolution results of the system (1) under the conditions specified in Experiment 6. The results indicate that altering system parameters in the P2P dataset significantly influences rumor

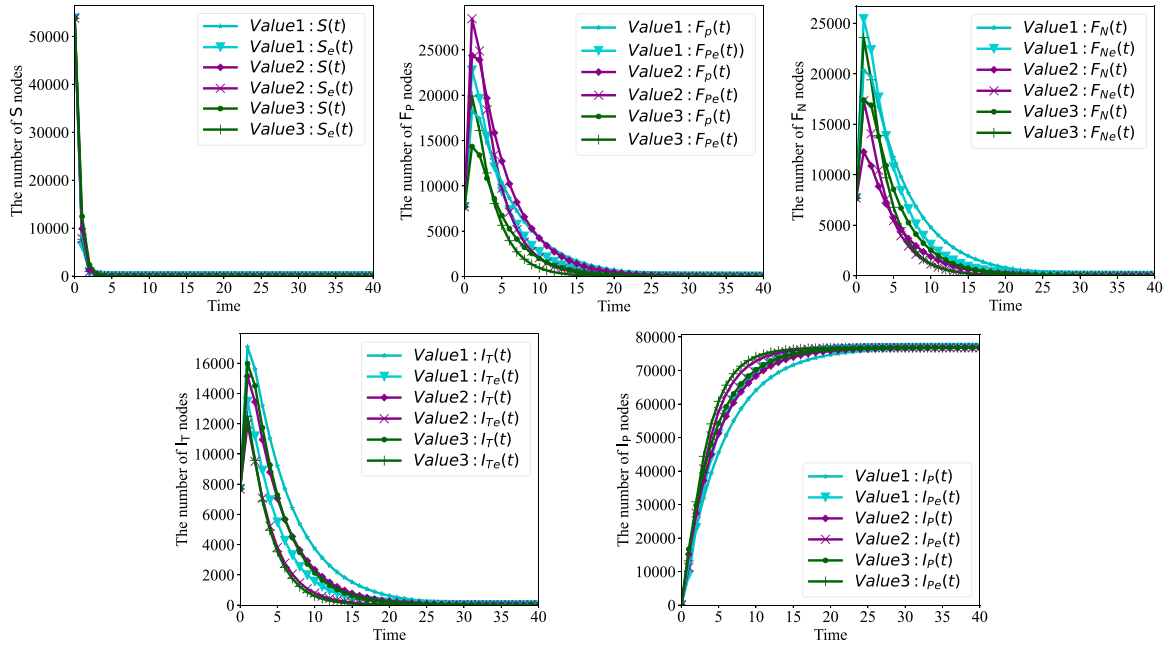


Fig. 6. Evolution of system (1) under given conditions in Experiment 4.

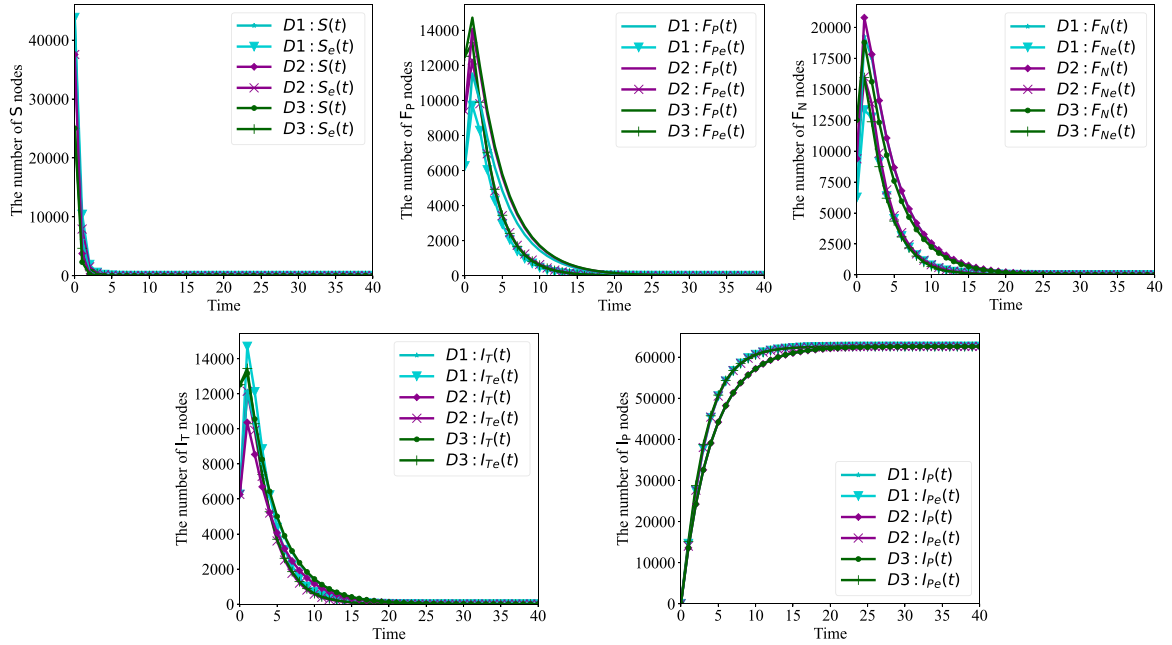


Fig. 7. Evolution of system (1) under given conditions in Experiment 6.

Table 8

System parameters in Experiment 6.

Parameter	β_1	β_2	ω_1	ω_2	α	η
Value1	0.33×10^{-4}	0.30×10^{-4}	0.25×10^{-2}	0.40×10^{-2}	0.5	0.3
Value2	0.26×10^{-4}	0.28×10^{-4}	0.30×10^{-2}	0.340×10^{-2}	0.4	0.35
Value3	0.36×10^{-4}	0.25×10^{-4}	0.44×10^{-2}	0.58×10^{-2}	0.45	0.3

propagation, consistent with the conclusions drawn in Experiments 2 and 4.

5.3. Model prediction verification using real rumor events

This subsection uses 13 real rumor datasets to verify the proposed model. All rumors in the first dataset (R1–R12) are based on emerging

rumors of Twitter, while the second dataset does not mention which social network the rumors are extracted from. In all cases, we use real network simulation verification of Twitter when we choose real network verification.

Experiment 7: In the NERT dataset, the feature marked as “a” represents the state of anti-rumor posts, which we believe that these users will no longer forward the rumor, corresponding to the I_P status in the proposed model. Due to the different feature situations of various rumor events, we select part of the data with the largest volume of rumors according to their respective features for preprocessing, as shown in Table 9, where there is no relevant data for the R4 event. Specifically, we statistically calculate the cumulative number of anti-rumor posts within an hour and use the aforementioned fitting method

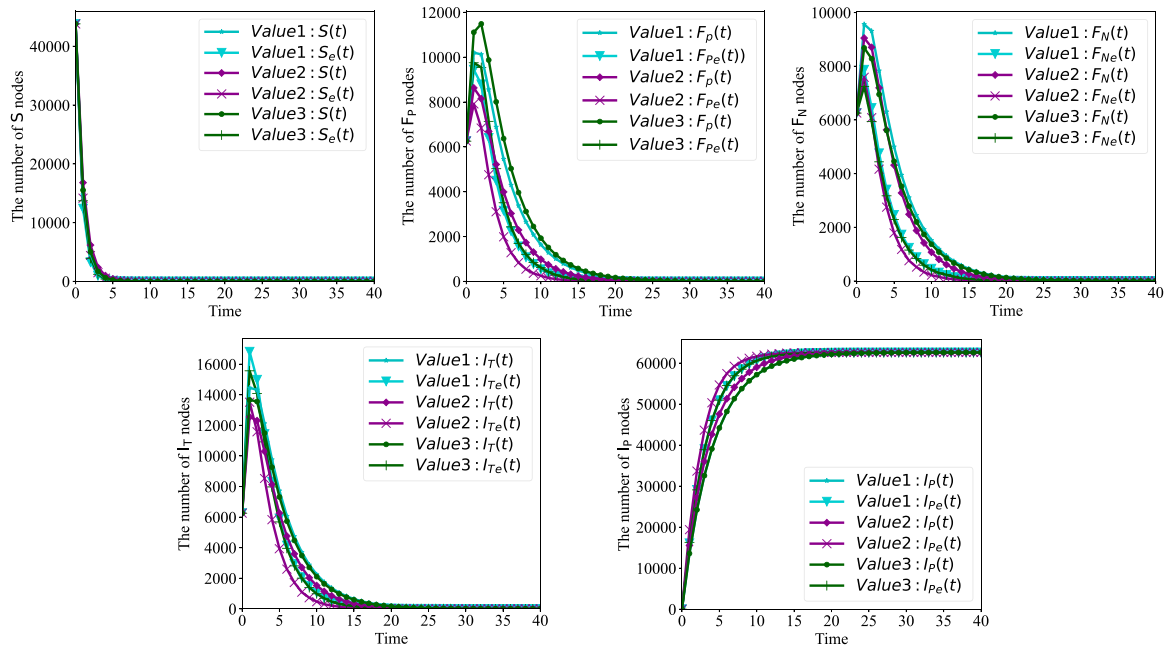


Fig. 8. Evolution of system (1) under given conditions in Experiment 6.

Table 9

Rumor characteristic situation.

Rumor incident	Start time	Duration (hours)
R1	At 15:00, 2017-12-13	48
R2	At 12:00, 2018-01-08	55
R3	At 04:00, 2018-01-08	81
R5	At 19:00, 2018-04-10	54
R6	At 23:00, 2018-06-28	42
R7	At 06:00, 2018-08-15	97
R8	At 19:00, 2018-09-06	29
R9	At 18:00, 2019-09-07	52
R10	At 18:00, 2017-09-07	63
R11	At 18:00, 2018-10-17	82
R12	At 16:00, 2018-10-26	76

Table 10

Fitting and validation time.

Rumor incident	Fitting time (hour)	Verification duration (hours)
R1	9	39
R2	39	16
R3	19	62
R5	29	25
R6	33	9
R7	89	8
R8	19	10
R9	39	13
R10	49	14
R11	69	13
R12	19	57

to fit the parameters while controlling the system parameters within the range of 0 to 1. We use the real Twitter network to simulate the rumor propagation process.

The duration of each rumor event is different. When fitting the data, we choose a relatively suitable proportion of the experiment to avoid the possibility of obtaining insufficient information with too little data or too much data, and the rumor tends to be stable and not persuasive. The specific proportion between the length of fitting time and verification time is shown in Table 10. Table 11 shows the parameters that this procedure fits.

Figs. 9–13 show the fitting, verification, and real network simulation prediction of rumor events with large data and long duration. The experimental results show that the proposed model can predict the spread trend of rumors after fitting. In the real network simulation prediction results, we can find that the number of users in each state tends to be stable after evolution. Satisfy Theorems 1–3.

Figs. 14–19 illustrate the fitting, validation, and real network simulation predictions for the remaining rumor events. Our findings indicate that the proposed model can generally predict the dissemination trends of these rumors. However, the simulations in the real network do not align with the anticipated evolution patterns. The rumors with poor prediction results, such as R7 and R8, R7 rumored that in August 2018, the children's show "VeggieTales" introduced a marijuana-themed character; its spread was limited due to parental caution and prompt refutation. R8 claimed that Michael Jordan resigned from Nike's board and took his Air Jordan line, threatening Nike's reputation, but the move was reversed after 29 h due to rapid suppression. Both rumors exhibited "self-limiting" traits, characterized by restricted propagation and limited attention. On Twitter, R7 and R8 had sparse retweet volumes, resulting in extreme parameter values during the fitting process. For instance, $\alpha \approx 1$ in R7 (almost all susceptible users become neutral), while $\alpha \approx 10^{-11}$ in R8 (almost none do). This is due to a few records of permanently immune users, causing the fitting algorithm to adjust parameters in ways that may violate their actual meanings. To better track such rumors, future research will add parameter constraints in fitting simulations to prevent extreme values. For data-sparse events, external interventions and modified fitting methods will be employed to prevent overfitting, thereby enhancing the model's ability to capture unique dynamics and improve prediction accuracy.

Experiment 8: For the rumor dataset with emotional labels, we calculated each piece of information's emotional polarity (positive and negative) according to the emotional values in the emotional labels, corresponding to the sum states in the proposed model. The r21 and r55 rumors are 19 months and 18 months, respectively. The first 10 months and the first 15 months are selected for parameter fitting, and the fitting parameters are shown in Table 10.

Figs. 20 and 21 show the experimental results of fitting and verifying rumors with emotional labels and simulating and predicting real networks. The experiment shows that the proposed model can be

Table 11
Fitted system parameters.

Rumor incident	β_1	β_2	ω_1	ω_2	α	η
R1	0.10087765971784259	0.1008742483605772	0.020904894604692	0.0028032947412179538	0.17218253738901976	0.15085447734019677
R2	1.0	1.0	0.0	0.0	1.0	0.00011980046103350406
R3	0.000146250342095939	0.002691034246302264	0.008207592886622681	0.00456483455222373	0.06005666288358989	0.05193335385406434
R5	0.3466784908761801	0.332701461634623	0.037832368284080445	0.02213309621377171	0.24800844802221464	0.07858414641578698
R6	$5.122594030528976 \times 10^{-5}$	$4.3457215181109165 \times 10^{-5}$	0.05799999006758581	0.005278361807591892	0.062098403953825446	0.031049230411429277
R7	0.01	0.00999999899777329	0.9999999045873573	0.9999999018192013	0.999999939020796	$1.9699769123253908 \times 10^{-5}$
R8	0.00592501035989353	1.0	0.999999999880835	0.0	$1.3382670755371472 \times 10^{-11}$	$2.278900740272012 \times 10^{-11}$
R9	0.0	0.0	0.0579999999915617	1.0	0.999999999999931	$5.6028604311933136 \times 10^{-5}$
R10	0.0	0.0	0.0	0.0	$2.654910090445314 \times 10^{-13}$	0.00044206420673700175
R11	$2.0032308675710318 \times 10^{-12}$	0.9999999943985517	0.9999999947220215	0.9999999945875521	0.999999966380694	0.0002165481428269956
R12	0.004592151581642084	0.004559292849939519	0.4729004178925963	0.019024869980451713	0.22382021488121942	0.11191728930932812
r21	0.00026799505236994956	0.0011422224973183254	0.00557356688523615	0.002434864673449366	0.5879374721549397	0.21751394053084153
r55	0.0002747080266184021	0.0	0.0	0.030268855587702453	0.11837164144507438	0.17009970235258595

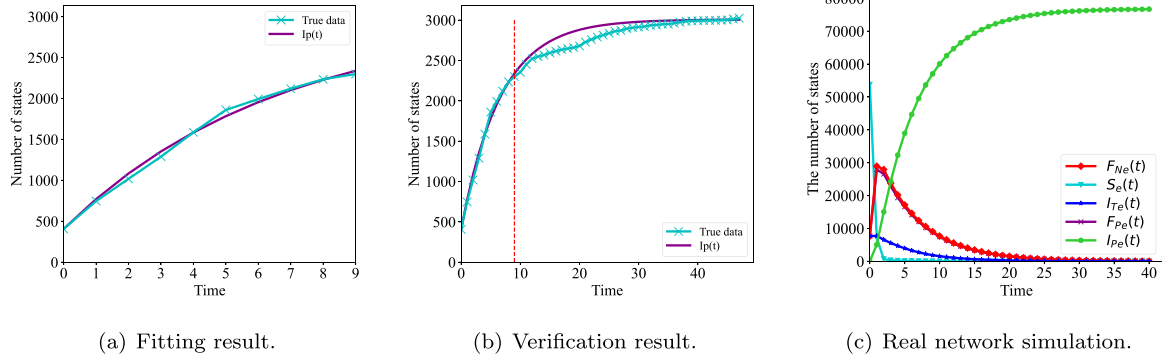


Fig. 9. Confirmation result of the R1 event (Figs. (a) and (b) for I_P state, Fig. (c) for all states).

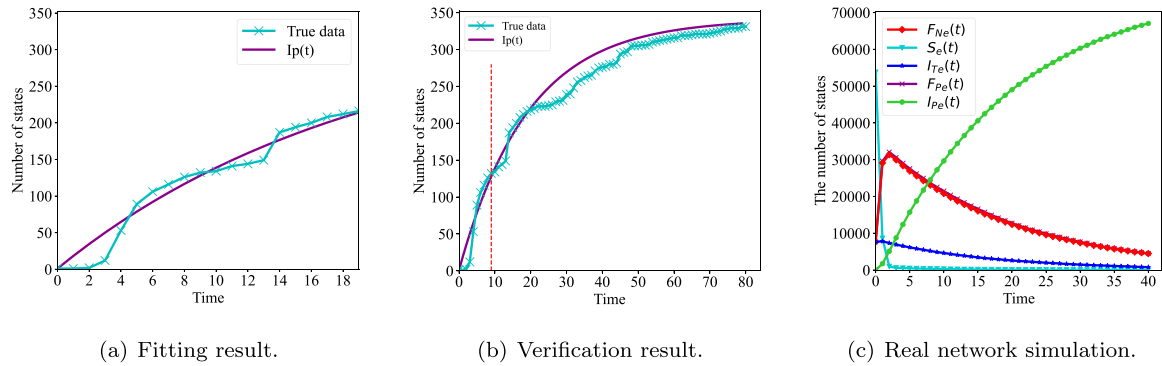


Fig. 10. Confirmation result of the R3 event (Figs. (a) and (b) for I_P state, Fig. (c) for all states).

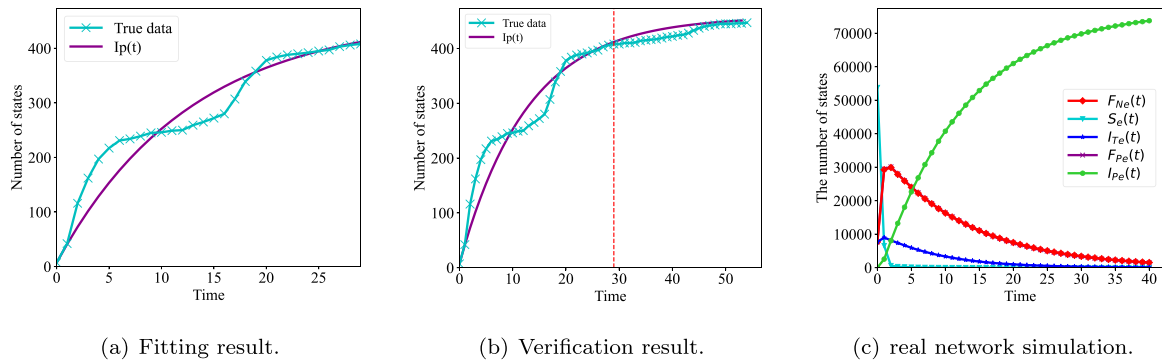


Fig. 11. Confirmation result of the R5 event (Figs. (a) and (b) for I_P state, Fig. (c) for all states).

consistent with the development trend of real events in the fitting time and can well predict the propagation trend of future events of rumors in the remaining time, and the system parameters fitted by the experiment

are simulated in the real network. The number of users in each state gradually becomes stable after evolution, satisfying [Theorems 1–3](#).

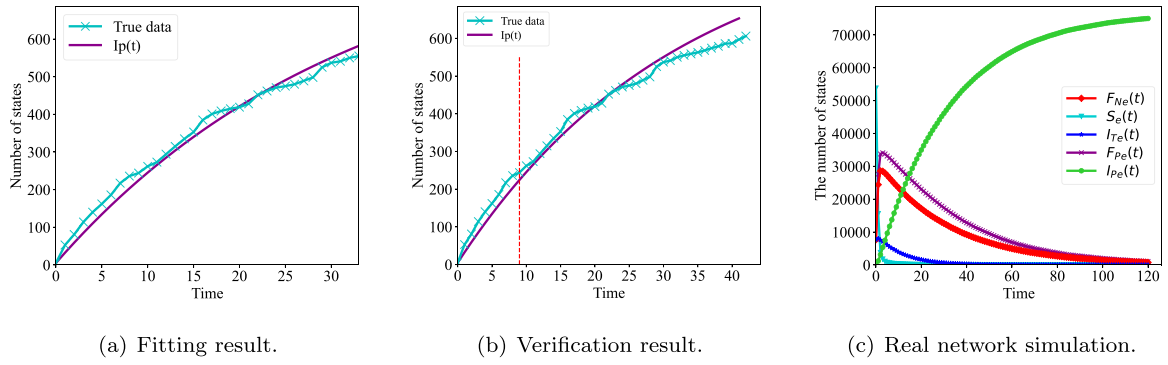


Fig. 12. Confirmation result of the R6 event (Figs. (a) and (b) for I_p state, Fig. (c) for all states).

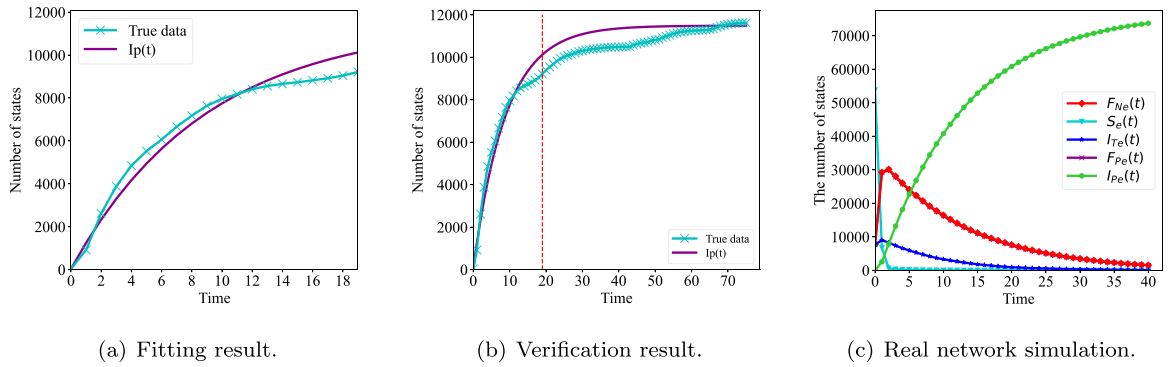


Fig. 13. Confirmation result of the R12 event (Figs. (a) and (b) for I_p state, Fig. (c) for all states).

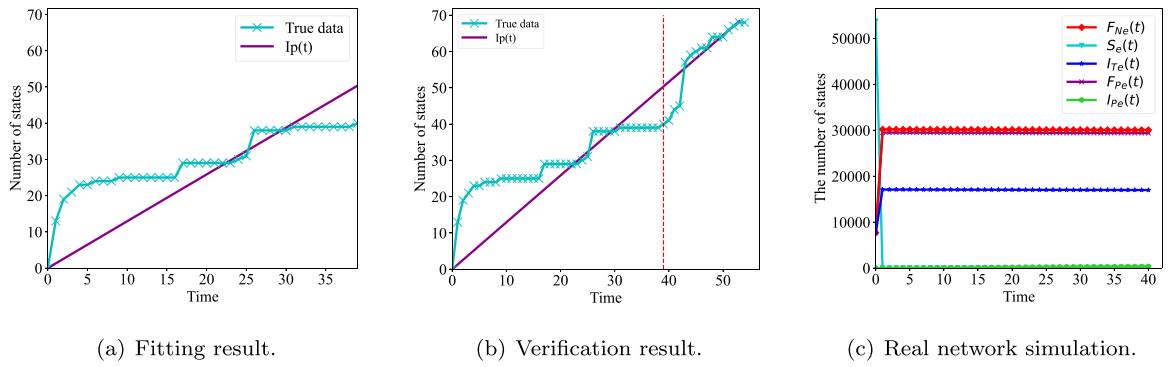


Fig. 14. Confirmation result of the R2 event (Figs. (a) and (b) for I_p state, Fig. (c) for all states).

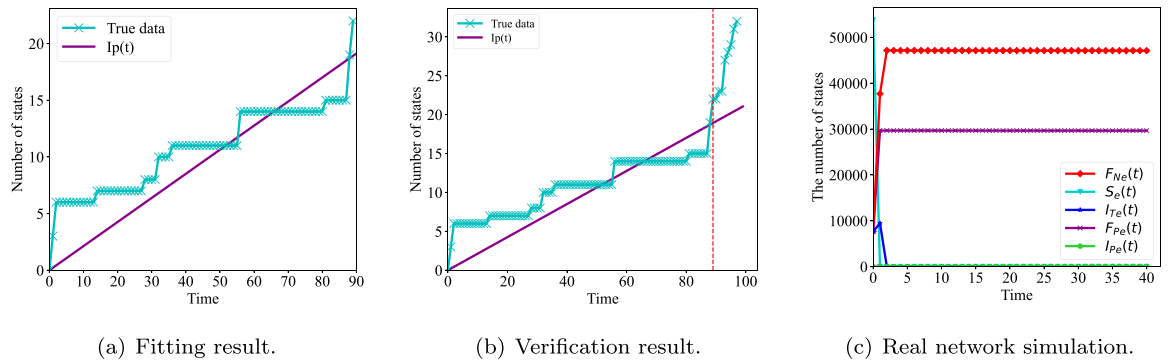


Fig. 15. Confirmation result of the R7 event (Figs. (a) and (b) for I_p state, Fig. (c) for all states).

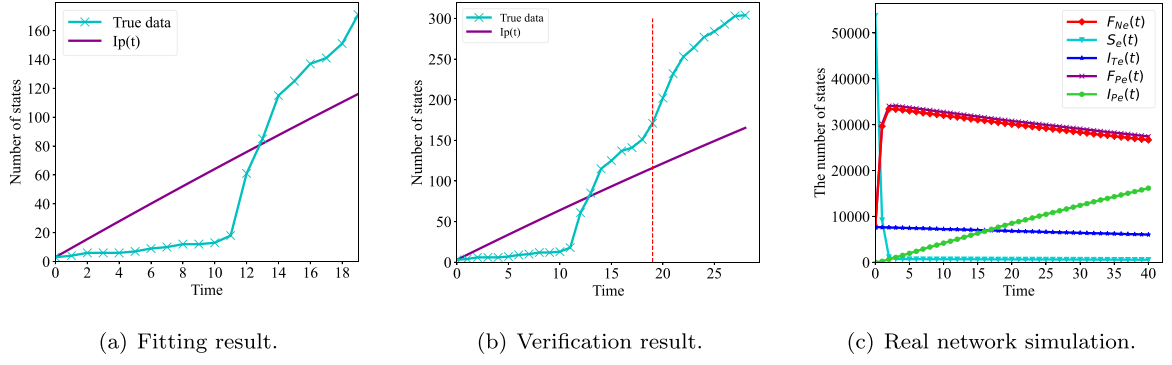


Fig. 16. Confirmation result of the R8 event (Figs. (a) and (b) for I_p state, Fig. (c) for all states).

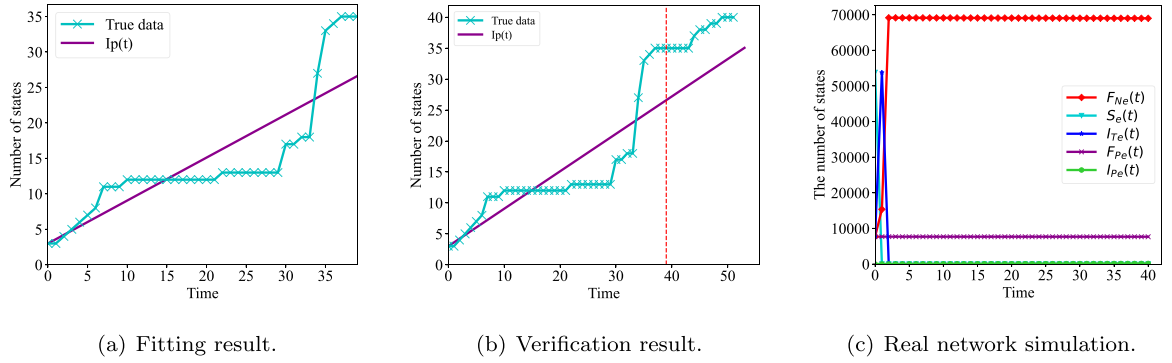


Fig. 17. Confirmation result of the R9 event (Figs. (a) and (b) for I_p state, Fig. (c) for all states).

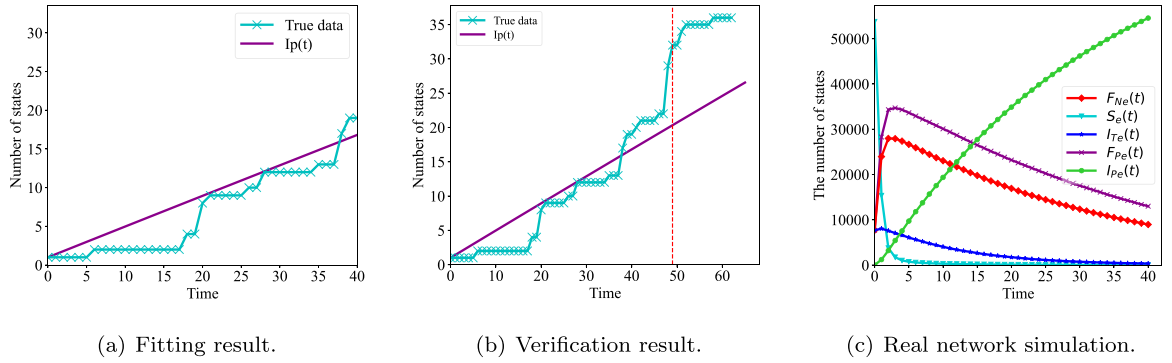


Fig. 18. Confirmation result of the R10 event (Figs. (a) and (b) for I_p state, Fig. (c) for all states).

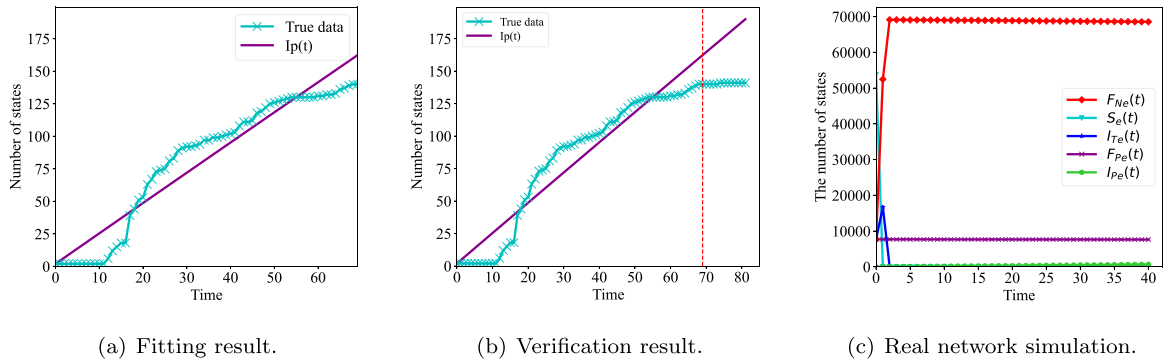


Fig. 19. Confirmation result of the R11 event (Figs. (a) and (b) for I_p state, Fig. (c) for all states).

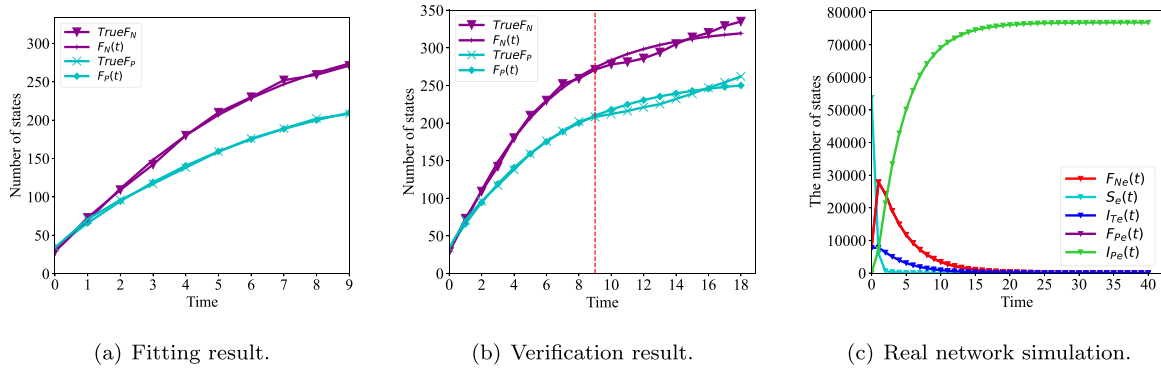


Fig. 20. Confirmation result of the r21 event (Figs. (a) and (b) for F_P and F_N states, Fig. (c) for all states).

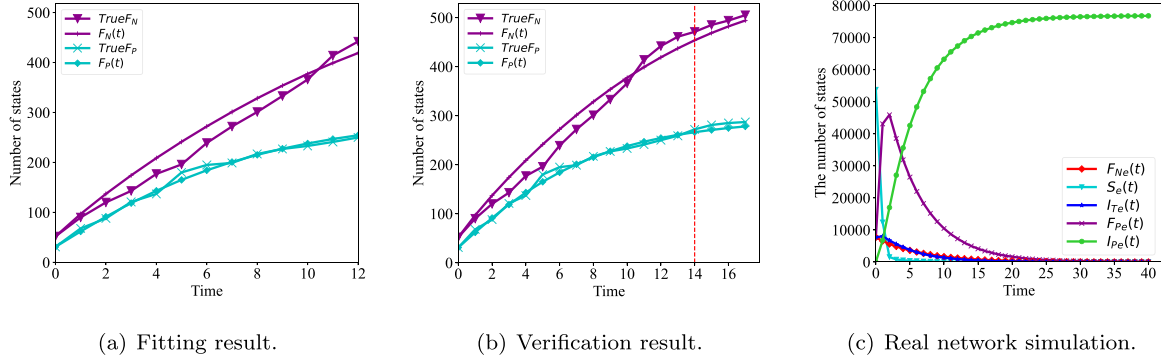


Fig. 21. Confirmation result of the r55 event (Figs. (a) and (b) for F_P and F_N states, Fig. (c) for all states).

Table 12
SFI model fitting system parameters.

Rumor incident	β	P	α
R1	0.0004176334527365764	0.5402590063793196	0.17218253738901976
R12	$7.117870320856934 \times 10^{-5}$	0.2986025796377154	0.2048948680963718
r21	0.0064291409140245165	0.09657157525444846	0.24064313046868657

Table 13
E-SFI model fitting system parameters.

Rumor incident	β	P_+	P_0	P_-	α_{Pos}	α_{Neu}	α_{Neg}
R1	0.0003561036702413848	0.19851534408507715	0.6879220702175155	0.08130877821912573	0.6238350340039741	0.539112249290898	0.677266208990392
R12	$5.983828842180312 \times 10^{-5}$	0.3093847420892513	0.14524456213464976	0.5724872787074741	0.36518989596479956	0.5341120035686896	0.3502601624013878
r21	0.9292247972458273	0.09132748359600382	0.04305949274938806	0.0005042883367278886	0.3216336636777925	0.5311375682065488	0.18118681339426473

5.4. Comparative analysis with traditional and new models

To better reflect the effectiveness of the proposed model, the SFI model (Yin, Shao, & Wu, 2019) and E-SFI model (Yin et al., 2021) will be compared next, and the model performance will be demonstrated. The effectiveness of the proposed model will be verified through the observation model fitting and prediction of the real rumor events.

The SFI model defines the susceptible state (S_1), forwarding state (F), and permanent immune state (I_1). In contrast, the E-SFI model defines the susceptible state (S_2), positive forwarding state (F_{Pos}), neutral forwarding state (F_{Neu}), negative forwarding state (F_{Neg}), and permanent immune state (I_2). To control all models on the same comparison axis and ensure that the basic conditions are the same, it is necessary to establish a corresponding relationship between the users' states of all models. Based on the model definition, we look for the correspondence between states of each model, assuming that all susceptible states are the same, forwarding states are mutually corresponding, and permanent immune states are the same. We can

obtain the following equation relationships:

$$\begin{cases} S_1(t) = S_2(t) = S(t), \\ F(t) = F_{Pos}(t) + F_{Neg}(t) = F_P(t) + F_N(t), \\ F_{Pos}(t) = F_P(t), \\ F_{Neg}(t) = F_N(t), \\ I_1(t) = I_2(t) = I(t). \end{cases} \quad (24)$$

Experiment 9: The number of neutral users in the E-SFI model is defined as 0, which is convenient for subsequent comparison. According to the real rumor events R1, R12, and r21 introduced in Section 5.3, we fit the comparison model similarly. The fitting time and verification time, as well as the corresponding event handling methods, are the same as those in Table 9 and Experiment 8 of Section 5.3. The parameters obtained by fitting the comparison model are shown in Tables 12 and 13 below, and the parameters of the proposed model are the fitting parameters corresponding to the rumor events R1, R12, and r21 in Table 11 of Section 5.3.

Fig. 22 shows the fitting results of the three models to the real events. We found that, compared with the two models, our model is superior to the comparison model in fitting real events and predicting

the trend of event propagation. Furthermore, the proposed model is more accurate in predicting the whole development process of real Internet rumor propagation.

5.5. Parameter analysis

Through observation, we find that the system parameters impact rumors spread in social networks. In this subsection, we will discuss the key parameters. Due to the similarity of results across different datasets, for the sake of simplicity, only the results from the Facebook dataset are presented here.

Experiment 10: In *Scenario 1*, consider system (1), using the same initial conditions: $(S(0), F_P(0), F_N(0), I_T(0), I_P(0)) = (2798, 399, 399, 399, 0)$ fixed other system parameters: $(\omega_1, \omega_2, \alpha, \eta) = (0.00058, 0.00029, 0.55, 0.22)$, for $\beta_1 = (0.0.000048, 0.0001)$ and $\beta_2 = (0.0.000053, 0.0001)$, each value is selected successively and its influence on rumor propagation is discussed. As can be seen from Fig. 2 and the model hypothesis, the probability that the user changes from susceptible to forwarding user, controlling its change, can be considered to change the intensity of positive emotion and negative emotion in the OSNs.

Fig. 23 shows the evolution process of F_P and F_N users under different emotional intensity levels in a fixed system parameter and the same initial conditions. The experiment shows that the more positive and negative emotional intensity levels, the more F_P and F_N users will evolve, indicating that rumors will spread more widely. This shows that users' emotions play a key role in the spread of rumors, and the stronger their emotions, the greater their desire to forward rumors, and the larger the scale of rumor spread. Conversely, we can reduce the number of rumor forwarders by controlling emotions. Therefore, when a rumor erupts, we should soothe the emotions of netizens and guide them correctly to control the spread of rumors and reduce the harm caused by rumors.

This section validates the proposed E-SF2I rumor propagation model through multi-level experiments. The results fully demonstrate the model's advantages in theoretical reliability, prediction accuracy, and practical value in social network rumor governance. Experiments 1–6 are based on three typical real network datasets of Facebook, Twitter, and P2P. By changing the initial user status and system parameters, the E-SF2I model is tested. The stability of the model, which is theoretically proven by the Lyapunov theorem, is verified in actual network scenarios. Its prediction results are not affected by network composition or fluctuations in user behavior and can be reliably applied to various social network scenarios. Experiments 7–8 verify the model using 13 real rumor events. The results show that the model can effectively capture the dynamic interaction of emotions and dual immunity in real rumor propagation and can be used as a practical tool for rumor early warning. Experiment 9 compares the E-SF2I model with the SFI and E-SFI models in three typical events, R1, R12, and r21. The results show that the prediction accuracy of the E-SF2I model is significantly higher than that of the SFI model, which ignores emotional factors, and the E-SFI model, which does not consider temporary neutral behavior. Experiment 10 analyzes the role of core parameters and clarifies that the key intervention directions for rumor governance can effectively control the scale of rumor propagation and reduce its social harm.

6. Discussions

The E-SF2I model integrates emotional differentiation and dual immunity states, addressing a key gap overlooked by traditional models that rarely combine both. It constructs a time-varying differential dynamic system, thus providing a theoretical basis for understanding interactions between user emotions (positive/negative forwarding) and behavioral states (temporary/permanent immunity). Extending epidemiological models like SIR and SFI, it captures neutral behaviors and positive/negative emotional states, critical for explaining real-world rumor persistence and recurrence. Unlike previous studies

that ignored emotions or simplified immunity to a single state, this study quantifies how emotional polarization accelerates the propagation of rumors and how temporary immunity mediates state transitions, revealing emotional arousal and immune decay as co-drivers of rumor dynamics. This enriches the understanding of “emotion-behavior” coupling in rumor propagation. Experiments show that user emotional intensity significantly impacts the spread of rumors; practically, intervening in network users' emotions can quickly control the propagation of rumors.

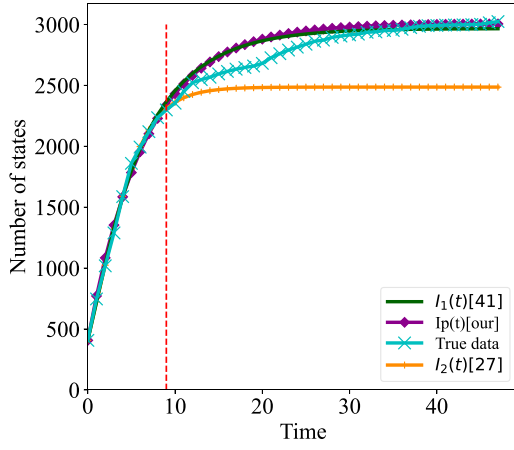
Through experiments, we found that this model can predict most real rumor events. In actual rumor management, this model can be used to monitor the trend of users' emotional forwarding in real time, quickly identify high-risk rumors, and intervene by reducing the visibility of emotional content. It can also track the probability of re-participation of temporarily immune users to evaluate the effectiveness of the intervention measures. Based on this model, phased intervention measures are implemented. In the early stage, the aim is to reduce the polarization of emotions. They can reduce the forwarding behavior of internet users through authoritative announcements, soothe the emotions of netizens, encourage temporarily immune users to become permanent immune users, and accelerate the quelling of rumors.

Specifically, social platforms can utilize the E-SF2I model to track the distribution of users across different states in real time, identify changes in these states (e.g., a surge in temporarily immune users or an increase in negative emotion forwarding). By capturing these state changes, leveraging rumor propagation and parameter fitting algorithms to obtain specific parameters, and simulating and predicting the overall rumor propagation trends, platforms can detect emerging rumor trends early, adjust content recommendation algorithms, reduce netizens' exposure to rumor content, and thereby inhibit the further spread of rumors. Organizations or enterprises needing to refute rumors can estimate the potential propagation scale based on the model, identify key time windows for response, and promptly carry out crisis management work. Similarly, they can share model-based predictions of rumor propagation with platforms, which helps predict geographical hotspots of rumor spread or the most severely affected groups. This enables targeted allocation of public communication resources before rumors escalate. In further research, attempts will be made to develop AI-driven early warning systems, combining this model with natural language processing methods to flag high-risk rumors before they spread, and implement real-time intervention strategies. The algorithm can immediately push refutational narratives to this group.

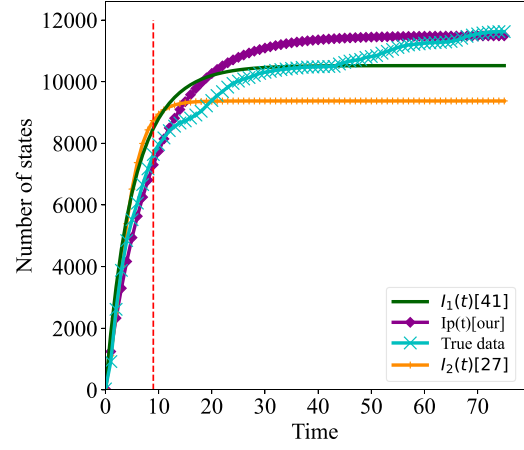
7. Conclusions

This paper conducted a series of studies considering the influence of emotional differentiation and dual immunity state coupling on rumor propagation. First, the E-SF2I rumor propagation dynamics model is proposed to describe the rumor propagation process on OSNs accurately. Then, using the Lyapunov stability theorem, the comprehensive effect of emotional differentiation and dual immunity state coupling on rumor propagation is analyzed from the theoretical level. To verify the reliability of the model, a multi-level verification method is constructed, and three real-world network datasets are used to verify the stability of the proposed model. In addition, 13 real rumor events were used for parameter fitting and prediction, which fully verified the validity of the model in real rumor events. Compared with several models, the superiority of the proposed model is highlighted. Finally, the parameter analysis reveals that the user emotion has a significant impact on the scale of rumor propagation and proposes that rumor propagation can be reduced by controlling the user emotion.

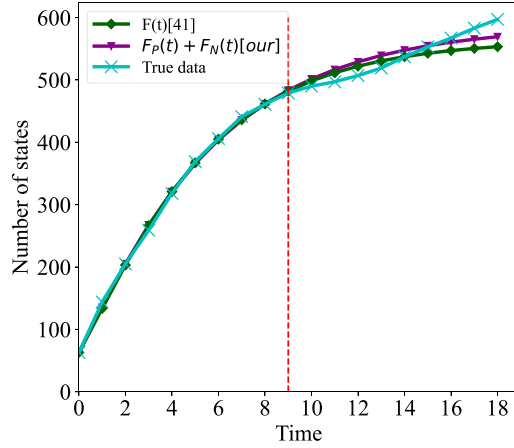
Although this study has made some achievements in constructing and verifying the rumor propagation model, there is still a broad space for exploration. In the future, the application scenarios of the model can be further expanded, and it can be extended to different types of social platforms and complex network structures to test the effectiveness of



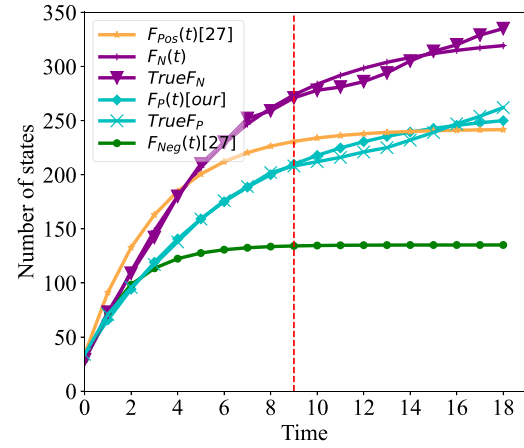
(a) Compare with two models using R1.



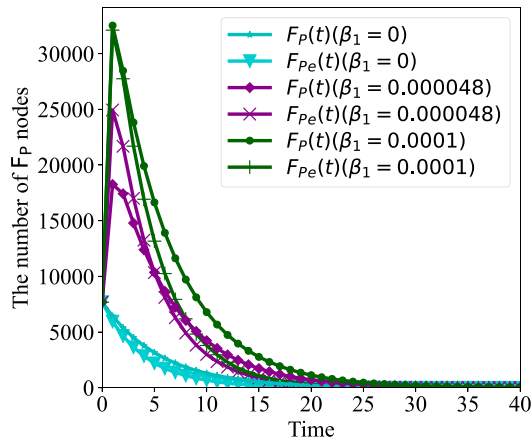
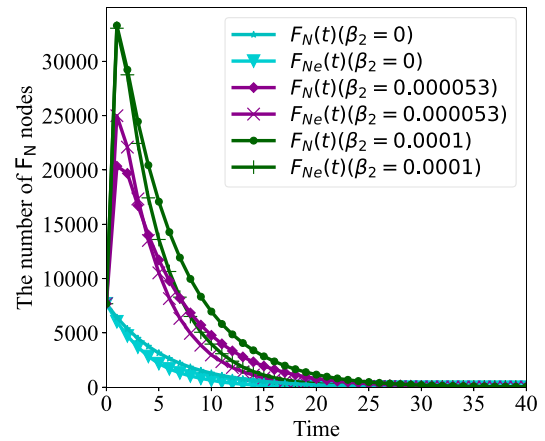
(b) Compare with two models using R12.



(c) Compare with the SFI model using r21.



(d) Compare with the E-SFI model using r21.

Fig. 22. Results of model comparison (Figs. (a) and (b) for I_p state, Figs. (c) and (d) for F_p and F_N states).(a) The influence of different β_1 .(b) The influence of different β_2 .**Fig. 23.** Influence of different positive and negative emotions intensity on forwarding rumors.

the proposed model. Then, combined with psychological theory and emotional infection theory, the mechanism of user emotion generation and change is deeply explored to regulate user emotion more accurately

to suppress rumor propagation, establish a more perfect rumor propagation model, and help formulate more targeted emotional intervention strategies. Since there is a multi-party game relationship between disseminators and rumor-tellers, government regulatory authorities and

rumor-makers, etc., in the process of rumor propagation, differential game theory can be introduced to study the control strategy of rumor propagation, determine the optimal supervision intensity and strategy of regulatory authorities under different circumstances, as well as the optimal communication strategy of rumor makers, to achieve effective control of rumor propagation.

CRedit authorship contribution statement

Chenquan Gan: Writing – review & editing, Writing – original draft, Methodology, Investigation. **Hongming Chen:** Writing – original draft, Validation, Software, Investigation. **Qingyi Zhu:** Supervision, Methodology. **Rui Tang:** Supervision, Formal analysis. **Deepak Kumar Jain:** Supervision, Formal analysis, Conceptualization. **Liping Feng:** Supervision, Methodology. **Salvador García:** Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors are grateful to the anonymous reviewers and the editor for their valuable comments and suggestions. This work was supported by the Guangxi Key Research and Development Program (No. AB24010317), the Fundamental Research Program of Shanxi Province (No. 202203021211116), and Jiangxi Provincial Key Laboratory of Electronic Data Control and Forensics (Jiangxi Police College) (No. 2025JXJYKFJJ002).

Data availability

Data will be made available on request.

References

- Afanasieva, L., Bukrieieva, I., Glyns'ka, L., Hlebova, N., & Falko, N. (2023). The communicative potential of multinational communities in countering destructive ethno-rumors in modern ukrainian society: a sociological analysis. *Cuestiones Políticas*, 40(75), 781–791.
- Bodaghi, A. (2019). Newly emerged rumors in Twitter. <http://dx.doi.org/10.5281/zenodo.2563864>, URL <https://doi.org/10.5281/zenodo.2563864>.
- Bodaghi, A., & Oliveira, J. (2020). The characteristics of rumor spreaders on Twitter: A quantitative analysis on real data. *Computer Communications*, 160, 674–687.
- Cai, Y., Yang, K., Huang, D., Zhou, Z., Lei, X., Xie, H., et al. (2019). A hybrid model for opinion mining based on domain sentiment dictionary. *International Journal of Machine Learning and Cybernetics*, 10, 2131–2142.
- Cao, D., Tang, X., Lin, Y., & Lin, D. (2025). Short video rumor detection based on causal graph. *Information Sciences*, 703, Article 121941.
- Chen, J., Ma, H., & Yang, S. (2023). SEIOR rumor propagation model considering hesitating mechanism and different rumor-refuting ways in complex networks. *Mathematics*, 11, 283.
- Fu, J., Song, Y., & Feng, Y. (2023). Rumor spreading model considering the roles of online social networks and information overload. *IEEE Access*, 11, 123947–123960.
- Gan, C., Yang, W., Zhu, Q., Li, M., Kumar Jain, D., Štruc, V., et al. (2025). Hybrid rumor debunking in online social networks: A differential game approach. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 55(4), 2513–2527.
- Gao, Y., Sun, Y., Zhang, L., Liu, F., & Gao, L. (2023). Identifying key rumor refuters on social media. *Expert Systems with Applications*, 231, Article 120603.
- Guo, H., & Yan, X. (2023). Dynamic modeling and simulation of rumor propagation based on the double refutation mechanism. *Information Sciences*, 630, 385–402.
- Hosni, A. I. E., Li, K., & Ahmad, S. (2020). Minimizing rumor influence in multiplex online social networks based on human individual and social behaviors. *Information Sciences*, 512, 1458–1480.
- Hu, X., Xiong, X., Wu, Y., Shi, M., Wei, P., & Ma, C. (2023). A hybrid clustered SFLA-PSO algorithm for optimizing the timely and real-time rumor refutations in online social networks. *Expert Systems with Applications*, 212, Article 118638.
- Jia, K., & Li, Z. (2020). Chinese micro-blog sentiment classification based on emotion dictionary and semantic rules. In *2020 international conference on computer information and big data applications* (pp. 309–312).
- Leskovec, J., Kleinberg, J., & Faloutsos, C. (2007). Graph evolution: Densification and shrinking diameters. *ACM Transactions on Knowledge Discovery from Data (TKDD)*, 1(1), 2–es.
- Li, M. Y., & Muldowney, J. S. (2000). Dynamics of differential equations on invariant manifolds. *Journal of Differential Equations*, 168(2), 295–320.
- Liqing, Q., & Shuqi, L. (2021). SVIR rumor spreading model considering individual vigilance awareness and emotion in social networks. *International Journal of Modern Physics C*, 32.
- Liu, X. (2024). Social network influence based on SHIR and SLPR propagation models. *Heliyon*, 10(18), Article e36658.
- Liu, X., He, D., Yang, L., & Liu, C. (2019). A novel negative feedback information dissemination model based on online social network. *Physica A: Statistical Mechanics and its Applications*, 513, 371–389.
- Mcauley, J., & Leskovec, J. (2012). Learning to discover social circles in ego networks. *NIPS*, 1, 539–547.
- Meng, L., Li, T., Huang, X., & Li, S. (2022). Lift the veil of rumors: The impact of the characteristics of information sources on the effectiveness of rumors spreading. *Internet Research*, 32(1), 90–119.
- Pan, W., Yan, W., Hu, Y., He, R., & Wu, L. (2023). Dynamic analysis of a SIDRW rumor propagation model considering the effect of media reports and rumor refuters. *Nonlinear Dynamics*, 111(4), 3925–3936.
- Pröllochs, N., Bär, D., & Feuerriegel, S. (2021). Emotions in online rumor diffusion. *EPJ Data Science*, 10(1), 51.
- Ripeanu, M., Foster, I., & Iamnitchi, A. (2002). Mapping the gnutella network: Properties of large-scale peer-to-peer systems and implications for system design. *ArXiv Preprint Cs/0209028*.
- Robinson, R. C. (2012). *An introduction to dynamical systems: continuous and discrete*. American Mathematical Soc..
- Shen, H., Tu, L., Guo, Y., & Chen, J. (2022). The influence of cross-platform and spread sources on emotional information spreading in the 2E-SIR two-layer network. *Chaos, Solitons & Fractals*, 165, Article 112801.
- Sun, T., Liu, C., Chen, L., Qian, Z., Li, P., & Zhu, Q. (2025). A unified framework for multi-modal rumor detection via multi-level dynamic interaction with evolving stances. *Information Processing & Management*, 62(3), Article 104066.
- Sun, X., Wang, Y., Chai, Y., & Liu, Y. (2024). Dynamic analysis and control strategies of the SEIHR rumor diffusion model in online social networks. *Applied Mathematical Modelling*, 134, 611–634.
- Tong, X., Jiang, H., Qiu, J., Luo, X., & Chen, S. (2023). Dynamic analysis of the IFCD rumor propagation model under stochastic disturbance on heterogeneous networks. *Chaos, Solitons & Fractals*, 173, Article 113637.
- Wan, P., Wang, X., Pang, G., Wang, L., & Min, G. (2023a). A novel rumor detection with multi-objective loss functions in online social networks. *Expert Systems with Applications*, 213, Article 119239.
- Wan, P., Wang, X., Pang, G., Wang, L., & Min, G. (2023b). A novel rumor detection with multi-objective loss functions in online social networks. *Expert Systems with Applications*, 213, Article 119239.
- Wang, R., Li, J., Sun, H., Duan, S., Zhang, G., & Xiao, Y. (2025). Rumor spreading model based on emotional characteristics and influence. *Expert Systems with Applications*, 274, Article 126952.
- Xiao, Y., & Chen, L. (2001). Analysis of a three species eco-epidemiological model. *Journal of Mathematical Analysis and Applications*, 258(2), 733–754.
- Yan, Y., Wang, Y., & Zheng, P. (2023). A graph-based pivotal semantic mining framework for rumor detection. *Engineering Applications of Artificial Intelligence*, 118, Article 105613.
- Yang, Z., Liu, B., Bi, X., Xiao, B., Li, W., Wang, G., et al. (2024). D-Net: A dual-encoder network for image splicing forgery detection and localization. *Pattern Recognition*, 155, Article 110727.
- Yin, M., Chen, W., Zhu, D., & Jiang, J. (2025). Enhancing video rumor detection through multimodal deep feature fusion with time-sync comments. *Information Processing & Management*, 62(1), Article 103935.
- Yin, F., Shao, X., & Wu, J. (2019). Nearcasting forwarding behaviors and information propagation in Chinese sina-microblog. *Mathematical Biosciences and Engineering*, 16, 5380–5394.
- Yin, F., Xia, X., Zhang, X., Zhang, M., Lv, J., & Wu, J. (2021). Modelling the dynamic emotional information propagation and guiding the public sentiment in the Chinese Sina-microblog. *Applied Mathematics and Computation*, 396, Article 125884.
- Yu, Z., Lu, S., Wang, D., & Li, Z. (2021). Modeling and analysis of rumor propagation in social networks. *Information Sciences*, 580, 857–873.
- Yu, Z., Wu, S., Zhang, Y., Cong, X., & Wu, K. (2024). IRLTS rumor propagation model considering the influence of lurking psychology and the truth of rumors. *Nonlinear Dynamics*, 112(20), 18593–18609.
- Yu, Z., Zi, H., Zhang, Y., Wu, S., Cong, X., & Mostafa, A. M. (2025). Dynamic modeling and simulation of double-rumor spreaders in online social networks with IS2TR model. *Nonlinear Dynamics*, 113(5), 4369–4393.
- Zheng, P., Dou, Y., & Yan, Y. (2024). Sensing the diversity of rumors: Rumor detection with hierarchical prototype contrastive learning. *Information Processing & Management*, 61(6), Article 103832.